Example: If $n \in I$, then $i^n \in \{1, -1, i, -i\}$.

***Complex Number*:** The number having the form $a+ib$ is called a complex number where $a, b \in R$. We can also define a complex number $a+ib$ in an ordered pair of real numbers (a, b) i, e. $a+ib=(a, b)$.

***Historical Note*:** The German mathematician Carl Friedrich Gauss (1777–1855), one of the greatest mathematician of the world, introduced the complex number system in 1832. The ordered pair system of complex number was introduced by the Irish mathematician William Rowan Hamilton (1805–65) in 1835.

6. *Real and Imaginary parts af a complex number;*

Let $z=(x, y)=x+iy$. Then x is called the ***real part*** of z

and it is denoted by Re (z) *i. e.* Re $(z) = x$. Again the ***imaginary part*** of z is y and it is denoted by Im (z) *i. e.* Im $(z) = y$.

where $n = 0, \pm 1, \pm 2, \pm 3, \quad \ldots \quad \ldots$

11 ***Equality of two complex numbers :***

(i) $a + ib = 0 \Rightarrow a = 0,\ b = 0,$

(ii) $a + ib = c + ib \Rightarrow a = c,\ b = d.$

Two complex numbers are said to be equal if and only if their real and imaginary parts are equal.

12. ***The distributive law :*** The distributive law holds in **C :** $\forall z_1, z_2, z_3 \in C,\ z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$. Here it is clear that $z_1(z_2 + z_3) = (z_2 + z_3) z_1$.

13. ***Absolute value or modulus :***

J. U. H. 86 ; R. U. H. 85.

Let $z = x + iy$ be a complex number, then the absolute value or modulus of z is denoted by $|z|$ and is given by

Final

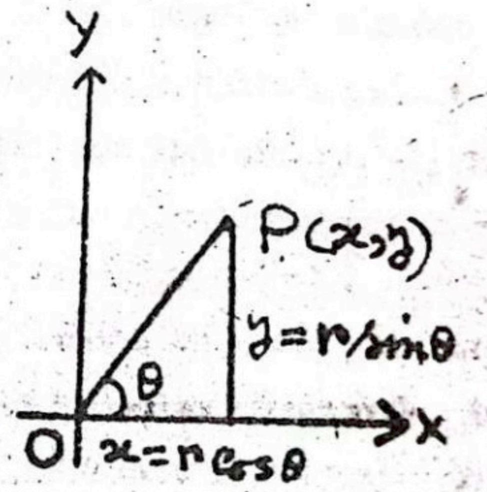
Final

$|z| = |x + iy| = \sqrt{(x^2 + y^2)}$. Here it is clear that $|z|$ is the distance from the origin and it is a non negative real number. ... $|3 + 4i| = \sqrt{(3^2 + 4^2)} = 5$, $|3 - 4i|$

6. *Polar (or trigonometric) form of complex numbers :*

Let $z=x+iy$ be a complex number which is represected by the vector ***OP***.

Let $OP=|\boldsymbol{OP}|=r$ and any angle θ (positive which the vector makes with the positive x–axis, then we have $x=r\cos\theta$, $y=r\sin\theta$,



Here $OP=|\boldsymbol{OP}|=\sqrt{(x^2+y^2)}=r$ is called the ***modulus*** or ***absolute value*** of $z=x+iy$ and it is denoted by $|z|$ or mod z. Again $\theta=\tan^{-1}y/x+2n\pi$, $n\in I$ is called the ***amplitude*** or ***argument*** or ***Phase*** of $z=x+iy$ and it is denoted by amp z or arg z. The form $z=x+iy=r\ (\cos\theta+i\sin\theta)$ is called the ***Polar form*** of the complex number, where r and θ is called the ***Polar coordinates***.

is called the ***Polar coordinates***.

(x) ***Modulus*** : ***J. U. H. 87 ; R. U. H. 85.***

Let (r, θ) be the polar coordinates corresponding to the complex number $z = x + iy = (x, y)$, then $r = \sqrt{(x^2 + y^2)}$ is called the ***modulus*** or ***absolute value*** of z where $r = |z| = \sqrt{(x^2 + y^2)}$.

21. Complex conjugate :

The complex conjugate of the number $z = x + iy$ is denoted by z and is defined by $\bar{z} = \overline{x + iy} = x - iy$.

85. Single-valued function :

A function $w = f(z)$ is called a single-valued in a domain S if only one value of w corresponds to each value of z in S.

86. Multiple-valued function :

R. U. H. 80.

A function $w = f(z)$ is called a multiple-valued in a domain S if more than one value of w corresponds to each value of z in S.

Any multiple valued function can be considered as a collection of single-valued functions where each single-valued member is called a ***branch*** of the function.

Example 64 : If $w = f(z) = z^2 + 2$, then w is called a single valued function of z since to each value of z there is only one value of w.

Example 65 : If $w = f(z) = z^{1/2}$, then w is called a multiple-valued function of z since to each value of z there are two values of w.

Example 66 : If $w = f(z) = z^2$, then $f(1+i) = (1+i)^2$
$= 1 + 2i + i^2 = 1 + 2i - 1 = 2i.$

ANALYTIC FUNCTIONS

134. Derivative or differential coefficient in a region or domain :

C. U. H. 83 ; D. U. H. T. 77. 85.

Let f(z) be a single-valued function defined in a region or domain R of the Argand plane, then the derivative of f(z) is defined as $f'(z) = \lim\limits_{\Delta z \to 0} \dfrac{f(z+\Delta z) - f(z)}{\Delta z}$ ... (1)

provided that the limit exists and is independent of the manner in which $\Delta z \to 0$. If the limit (1) exists, then f(z) is also called ***differentiable*** at z.

In the equation (1), sometimes we will use h instead of Δz.

135. Differentiable at a point :

D. U. H. T. 77, 85 ; D. U. H. 89, 90, 91 ; R. U. H. 86 ; J. U. H. 87 ; R. U. M. SC. P. 84 ; C. U. H. 83.

A single-valued function f(z) is said to be differentiable at a point $z = z_0$ of the Argand plane

if the limit $f'(z_0) = \lim\limits_{\Delta z \to 0} \dfrac{f(z_0+\Delta z) - f(z)}{\Delta z}$ ... (2) exists.

136. $\epsilon - \delta$ definition of the derivative at a point ;

The single-valued function f(z) has a derivative $f'(z_0)$ at $z = z_0$ of the Argand plane if given any $\epsilon > 0$, we can find $\delta > 0$ such that $\left| \dfrac{f(z_0+\Delta z) - f(z_0)}{\Delta z} - f'(z_0) \right| < \epsilon$

whenever $0 < |\Delta z| < \delta$.

Proof : The proof of this theorem is given in chapter 5.

142. *Analytic (or regular or holomorphic) function* :

C. U. H. T. 77, 85, 88, 91 ; C. U. H. 87, 88 ; R. U. H. 73, 75, 77, 82, 85 ; D.U. H. S. 85 ; R. U. M. SC. P. 84, 85, 86, 88; D. U. M. SC. P. T. 91 ; R. U. 80 ; D. U. M. SC. P. 78 ; C. U. H. 85, 89.

A single-valued function $f(z)$ is said to be analytic in a region R if the derivative $f'(z)$ exists at all points z of the region R.

143. *Analytic at a point* :

D. U. H. T. 85 ; R. U. M. SC. P. 85 ; D. U. H. 89, 91 ; J. U. H. 86, 87, 91.

A single-valued function $f(z)$ is said to be analytic at a point z_0 only if there is a neighbourhood $|z - z_0| < \delta$ at all points of which $f'(z)$ exists.

149. Cauchy-Riemann equations :

Necessary and sufficient conditions for the function $f(z)=u(x, y)+iv(x, y)$ to be analytic.

D. U. H. 83, 86, 89, 90 ; J. U. H. 85, 87 ; D. U. H. 86 ; D. U. 83 ; C. U. H. 82.

(i) Necessary condition for $f(z)$ to be analytic (Thaorem 72) : D. U. H. T. 84, 87 ; R. U. M. SC, P. 84, 85 ; R. U. H. 73, 77, 82, 90 ; D. U. H 87, 91 ; R. U. 78 ; D. U. M. SC. P. 89 ; C. U. H. 86, 87, 88, 90.

If $f(z)=u(x, y)+iv(x, y)$ is analytic in a region R, then it is necessary that the four partial derivatives $u_x = \frac{\delta u}{\delta x}$, $v_x = \frac{\delta v}{\delta x}$, $u_y = \frac{\delta u}{\delta y}$, $v_y = \frac{\delta v}{\delta y}$ should exist and satisfy the Cauchy-Riemann equations $u_x = v_y$, $u_y = -v_x$.

Proof : If $f(z) = u(x, y) + iv(x, y)$ is analytic, then

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{\{u(x+\Delta x, y+\Delta y) + iv(x+\Delta x, y+\Delta y)\} - \{u(x,y) + iv(x,y)\}}{\Delta x + i\Delta y} \quad \ldots \quad \ldots \quad \ldots (1).$$

must exists and is unique. In this case it is independent of the manner in which $\Delta z \to 0$ or $\Delta x \to 0$ and $\Delta y \to 0$. At z:

(a) Let $\Delta y = 0$ and $\Delta x \to 0$ $\Delta y = 0$ and $\Delta x \to 0$.

Then $(1) \Rightarrow f\ (z) = \lim_{\Delta x \to 0} \left\{ \frac{u(x+\Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x+\Delta x, y) - v(x, y)}{\Delta x} \right\}$

which implies that u_x and v_x must exist and the limit becomes $f'(z) = u_x + iv_x \quad \cdots \quad (2)$.

(b) Let $\Delta y = 0$ and $\Delta y \to 0$ $\Delta x = 0$ and $\Delta y \to 0$

Then $(1) \Rightarrow f'(z) = \text{lime}_{\Delta y \to 0} \left\{ \frac{n(x, y+\Delta y) - u(x, y)}{i\Delta y} + \frac{v(x, y+\Delta y) - v(x, y)}{\Delta y} \right\}$

which imPlies that u_y and v_y must exist and the limit becomes

$f'(z) = -iu_y + v_y \quad \cdots \quad (3)$

Since f(z) is analytic, then (2) and (3) are identical and we have $u_x + iv_x = -iu_y + v_y \Rightarrow u_x = v_y$, $u_y = -v_x$.

the German mathematician G. F. B. Riemann (1826—1866).

(ii) *Sufficient condition for* $f(z)$ *to be analytic* (Theorem 73) : R. U. M. SC. P. 86, 88 ; R. U. H. 75 ; J U.H. 86, 88, 86 . R. U 80 ; C. U. H. 85, 89.

The continuous single-valued function $f(z)=u(x, y)+iv(x, y)$ is analytic in a region R if the four partial derivatives u_x, v_x, u_y, v_y exist, are continuous and satisfy the Cauchy-Riemann equations $u_x=v_y$, $u_y=-v_x$ in R.

Proof : Since u_x and u_y are continuous, then we have

$$\Delta u = u(x+\Delta x, y+\Delta y)-u(x, y)$$

$$=\{u(x+\Delta x, y+\Delta y)-u(x, y+\Delta y)\}+\{u(x, y+\Delta y)-u(x,y)\}$$

$$=\left(\frac{\partial u}{\partial x}+\epsilon_1\right)\Delta x+\left(\frac{\partial u}{\partial y}+\eta_1\right)\Delta y \quad \left[\text{by the mean value theorem}\right]$$

$$=\frac{\partial u}{\partial x}\Delta x+\frac{\partial u}{\partial y}\Delta y+\epsilon_1\Delta x+\eta_1\Delta y \qquad (1)$$

where $\epsilon_1\to 0$ and $\eta_1\to 0$ as $\Delta x\to 0$ and $\Delta y\to 0$.

Similarly, since v_x and v_y are continuons then we have

$$\Delta v=\frac{\partial v}{\partial x}\Delta x+\frac{\partial v}{\partial v}\Delta y+\epsilon_2\Delta x+\eta_2\Delta y \quad \cdots \quad (2)$$

where $\epsilon_2\to 0$ and $\eta_2\to 0$ as $\Delta x\to 0$ and $\Delta y\to 0$.

Now adding (1) and (2) $\Rightarrow$ $\Delta w=\Delta u+i\Delta v$

$$=\left(\frac{\partial u}{\partial x}+i\frac{\partial v}{\partial x}\right)\Delta x+\left(\frac{\partial u}{\partial y}+i\frac{\partial v}{\partial y}\right)\Delta y+\epsilon\Delta x+\eta\Delta y \cdots \quad (3)$$

where $\epsilon=\epsilon_1+i\epsilon_2\to 0$ and $\eta=\eta_1+i\eta_2\to 0$ as $\Delta x\to 0$ aud $\Delta y\to 0$

Now by the Cauchy-Riemann equations (3) $\Rightarrow$

$$\Delta w=\left(\frac{\delta u}{\delta x}+i\frac{\delta v}{\delta x}\right)\Delta x+\left(-\frac{\delta v}{\delta x}+i\frac{\delta u}{\delta x}\right)\Delta y+\epsilon\Delta x+\eta\Delta y$$

$$=\left(\frac{\delta u}{\delta x}+i\frac{\delta v}{\delta x}\right)\Delta x+i\left(\frac{\delta u}{\delta x}+i\frac{\delta v}{\delta x}\right)\Delta y+\epsilon\Delta x+\eta\Delta y$$

$$=\left(\frac{\delta u}{\delta x}+i\frac{\delta v}{\delta x}\right)(\Delta x+i\Delta y)+\epsilon\Delta x+\eta\Delta y$$

$$=\left(\frac{\delta u}{\delta x}+i\frac{\delta v}{\delta x}\right)\Delta z+\epsilon\Delta x+\eta\Delta y \quad \cdots \quad (4)$$

Now dividing (4) by Δz and taking the limit as $\Delta z\to 0$, we have $\frac{dw}{dz}=f'(z)=\lim_{\Delta z\to 0}\frac{\Delta w}{\Delta z}=\frac{\delta u}{\delta x}+i\frac{\delta v}{\delta x}$. so that the derivative exists and is unique and therefore $f(z)$ is analytic in R.

153 Cauchy-Riemann equations in polar form :

Theorem 77 : Show that Cauchy-Riemann equations in polar form are :

$$\frac{\delta u}{\delta r}=\frac{1}{r}\frac{\delta v}{\delta \theta},\quad \frac{\delta v}{\delta r}=-\frac{1}{r}\frac{\delta u}{\delta \theta}$$

D. U. H. T. 77 ; J. U. H. 84 ; D. U. M. SC. P. 89 ; C. U. H. 83, 88.

Prnof : We have $w=u+iv$ ... (1) and $z=x+iy$ $=r(\cos\theta+i\sin\theta)=re^{i\theta}$ ... ... (2).

Let $\left(\frac{dw}{dz}\right)_\theta$ denotes the value of $\frac{dw}{dz}$ when $\theta=$ Constant and $\left(\frac{dw}{dz}\right)_r$, the value of $\frac{dw}{dz}$ when $r=$constant.

Now if w is analytic, then $\frac{dw}{dz}=\frac{\delta w}{\delta r}\left(\frac{dr}{dz}\right)_\theta$ ... (3)

and $\frac{dw}{dz}=\frac{\delta w}{\delta \theta}\left(\frac{d\theta}{dz}\right)_r$ ... (4)

Now (2) $\Rightarrow r=z\,e^{-i\theta}\Rightarrow\left(\frac{dr}{dz}\right)_0=e^{-i\theta}$ ... ... (5). Then by (3) aud (5) we have $\frac{dw}{dz}=\frac{\delta w}{\delta r}\,e^{-i\theta}$ ... ... (6).

Again (2) $\Rightarrow e^{i\theta}=\frac{z}{r}\Rightarrow ie^{i\theta}\left(\frac{d\theta}{dz}\right)_r=\frac{1}{r}\Rightarrow$ $\left(\frac{d\theta}{dz}\right)_r=-\frac{i}{r}e^{-i\theta}$ .. ... (7). Then by (4) and (7) we have $\frac{dw}{dz}=-\frac{i}{r}\frac{\delta w}{\delta \theta}\,e^{-i\theta}$ ... ... (8)

Now by (6) and (8) we get $\frac{\delta w}{\delta r}\,e^{-i\theta}=-\frac{i}{r}\frac{\delta w}{\delta \theta}e^{-i\theta}$

$\Rightarrow\frac{\delta w}{\delta r}=-\frac{i}{r}\frac{\delta w}{\delta \theta}\Rightarrow\frac{\delta u}{\delta r}+i\frac{\delta v}{\delta r}=-\frac{i}{r}\left(\frac{\delta u}{\delta \theta}+i\frac{\delta v}{\delta \theta}\right)\Rightarrow$

$\frac{\delta u}{\delta r}=\frac{1}{r}\frac{\delta v}{\delta \theta},\ \frac{\delta v}{\delta r}=-\frac{1}{r}\frac{\delta u}{\delta \theta}$ and the theorem is proved.

Example 95 : Verify the Cauchy-Riemann equations are satisfied for the following functions :

(*i*) $f(z)=e^{z^2}$. *D. U.H. 88.* and

(*ii*) $f(z)=\cos 2z$. *D. U. H. T. 83.*

Solution (*i*) : We have $f(z)=u+iv=e^{(x+iy)^2}$

$$=e^{x^2-y^2+2xyi}=e^{x^2-y^2}(\cos 2xy+i\sin 2xy).$$

Then $u=e^{x^2-y^2}\cos 2xy$ and $v=e^{x^2-y^2}\sin 2xy$.

$$\Rightarrow \frac{\partial u}{\partial x}=e^{x^2-y^2}(2x\cos 2xy-2y\sin 2xy)=\frac{\partial v}{\partial y} \text{ and}$$

$$\frac{\partial u}{\partial y}=-e^{x^2-y^2}(2y\cos 2xy+2x\sin 2x)=-\frac{\partial v}{\partial x}.$$

Thus u and v satisfy the Cauchy-Riemann equations.

(*ii*) We have $f(z)=u+iv=\cos(2x+2yi)=\cos 2x\cosh 2y-i\sin 2x\sinh 2y \Rightarrow u=\cos 2x\cosh 2y$ and $v=-\sin 2x\sinh 2y$

$$\Rightarrow \frac{\partial u}{\partial x}=-2\sin 2x\cosh 2y=\frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y}=2\cos 2x\sinh 2y$$

$=-\frac{\partial v}{\partial x}$ Thus u and v satisfy the Cauchy-Riemann equations.

161. *Laplace's equation in two dimensions* :

In two dimensions a second order partial differential equation $\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0$ or $\nabla^2\phi=0$ is called Laplace's equation. The vector operator ∇^2 is called the ***Laplacian*** where $\nabla^2=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}$

162. *Harmonic function* :

D. U. H. 86, 87, 89, 91 ; D. U. H. T. 83, 88, 90, 91 ; D. U. H. S. 85 ; J. U. H. 87, 91 ; D, U. M. SC. P. T. 91 ; C. U. M. SC. P. 78, 89 ; C. U. H. 90

A real-valued function $\phi(x, y)$ is called a harmonic function or simply harmonic in a region R if all its second-order partial derivatives are continuous in R and at each point of the region it satisfies the Laplace's equation $\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0.$

163. *Conjugate harmonic function* :

D. U. H. 86, 88 ; D. U, H. T. 90, 91 ; J. U. H. 88, 91 ; D. U. M. SC. P. 89.

If $f(z)=u+iv$ is an analytic function, then u and v are called conjugate harmonic functions or harmonic conjugate functions or simply conjugate functions.

N. B. Here v is called the conjugate harmonic function of u and u is called the conjugate harmonic function of $-v$.

165. Laplace's equation in polar form :

Theorem 85 : Show that the Laplace's equation in polar form is : $\frac{\partial^2\psi}{\partial r^2}+\frac{1}{r}\frac{\partial\psi}{\partial r}+\frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2}=0.$

D. U. M. SC. P. 89 ; J. U. H. 91.

***proof* :** The Cauchy-Riemann equations in polar form are :

$$\frac{\partial u}{\partial r}=\frac{1}{r}\frac{\partial v}{\partial\theta} \text{ or } \frac{\partial v}{\partial\theta}=r\frac{\partial u}{\partial r} \quad \cdots \quad \cdots \quad (1) \text{ and}$$

$$\frac{\partial v}{\partial r}=-\frac{1}{r}\frac{\partial u}{\partial\theta} \text{ or } \frac{\partial u}{\partial\theta}=-r\frac{\partial v}{\partial r} \quad \cdots \quad \cdots \quad (2)$$

If the second partial derivatives of u and v with respect to r and θ are continuous, then

$$\frac{\partial^2 u}{\partial r\partial\theta}=\frac{\partial^2 u}{\partial\theta\partial r} \quad \cdots \quad \cdots \quad (3) \text{ and } \frac{\partial^2 v}{\partial r\partial\theta}=\frac{\partial^2 v}{\partial\theta\partial r} \quad \cdots \quad \cdots \quad (4)$$

Now, by (1) and (2), (4) $\Rightarrow \frac{\partial}{\partial r}\left(\frac{\partial v}{\partial\theta}\right)=\frac{\partial}{\partial\theta}\left(\frac{\partial v}{\partial r}\right)$

$$\Rightarrow \frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)=\frac{\partial}{\partial\theta}\left(-\frac{1}{r}\frac{\partial u}{\partial\theta}\right) \Rightarrow r\frac{\partial^2 u}{\partial r^2}+\frac{\partial u}{\partial r}=-\frac{1}{r}\frac{\partial^2 u}{\partial\theta^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}+\frac{1}{r^2}\frac{\partial^2 u}{\partial\theta^2}=0 \quad \cdots \quad (5).$$

Again by (1) and (2), (3) $\Rightarrow \frac{\partial}{\partial r}\left(\frac{\partial u}{\partial\theta}\right)=\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial r}\right)$

$$\Rightarrow \frac{\partial}{\partial r}\left(-r\frac{\partial v}{\partial r}\right)=\frac{\partial}{\partial\theta}\left(\frac{1}{r}\frac{\partial v}{\partial\theta}\right) \Rightarrow -r\frac{\partial^2 v}{\partial r^2}-\frac{\partial v}{\partial r}=\frac{1}{r}\frac{\partial^2 v}{\partial\theta^2}$$

$\Rightarrow \frac{\partial^2 v}{\partial r^2}+\frac{1}{r}\frac{\partial v}{\partial r}+\frac{1}{r^2}\frac{\partial^2 v}{\partial\theta^2}=0 \quad \cdots \quad (6).$ Thus (5) and (6) $\Rightarrow \frac{\partial^2\psi}{\partial r^2}+\frac{1}{r}\frac{\partial\psi}{\partial r}+\frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2}=0,$ which is called Laplace's equation in polar form.

$\ldots y + 6xy^2 - y^3 - 2x^3 + c.$

Example 109 : Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is harmonic function. Find v such that $u + iv$ is analytic.

R. U. H. 86 ; C. U. H. 86.

Solution : We have $\dfrac{\partial u}{\partial x} = 3x^2 - 3y^2 + 6x$ ... ... (1).

$\dfrac{\partial^2 u}{\partial x^2} = 6x + 6$ ... (2), $\dfrac{\partial u}{\partial y} = -6xy - 6y$ ... ... (3) and

$\dfrac{\partial^2 u}{\partial y^2} = -6x - 6$ ... (4). Now $(2)+(4) \Rightarrow \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$

$\Rightarrow u$ is harmonic function. Again, by Cauchy—Riemenn equations we have $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y} = 3x^2 - 3y^2 + 6x$ ... (5)

and $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x} = 6xy + 6y$ ... (6).

Now integrating (5) with respect to $y \Rightarrow v = 3x^2y - y^3 + 6xy + F(x)$ ... (7). Now by (6) and (7) $\Rightarrow 6xy + 6y + F'(x) = 6xy + 6y \Rightarrow F'(x) = 0 \Rightarrow F(x) = c$ ... (8). Now by (7) and (8) $\Rightarrow v = 3x^2y - y^3 + 6xy + c.$

Show that $\psi(x, y) = \frac{1}{2}\log(x^2 + y^2)$ is a

171. *Construction of an analytic function:*

Theorem 91: (Milne Thomson Method or simply Milne Method)

If $f(z)=u(x,y)+iv(x,y)$ is analytic and if $\frac{\delta u}{\delta x}=\phi_1(x,y)$,

$\frac{\delta u}{\delta y}=\phi_2(x,y)$ and $\frac{\delta v}{\delta x}=\psi_1(x,y)$, $\frac{\delta v}{\delta y}=\psi_2(x,y)$, then

(i) $f'(z)=\phi_1(z,0)-i\,\phi_2(z,0)$;

(ii) $f(z)=\int[\phi_1(z,0)-i\,\phi_2(z,0)]\,dz+c$;

(iii) $f'(z)=\psi_1(z,0)+i\,\psi_2(z,0)$;

(iv) $f(z)=\int[\psi_1(z,0+i\psi_2(z,0))]\,dz+c$.

Proof: *(i)* We have $f'(z)=\frac{\delta u}{\delta x}+i\frac{\delta v}{\delta x}=\frac{\delta u}{\delta x}-i\frac{\delta u}{\delta y}$,

by Cauchy-Riemann equations $\Rightarrow f'(z)=\phi_1(x,y)-i\phi_2(x,y)$ ... (1)

Now putting $y=0$ in (1) $\Rightarrow f'(x)=\phi_1(x,0)-i\,\phi_2(x,0)$ ... (2)

Again replacing x by z, (2) $\Rightarrow f'(z)=\phi_1(z,0)-i\,\phi_2(z,0)$ ... (3)

(ii) Now integrating (3) with respect to $z \Rightarrow$

$f(z)=\int[\phi_1(z,0)-i\,\phi_2(z,0)]+c$.

(iii) and *(iv)*: Try yourself.

172 Theorem 92: If f(z)...

Example 117 : Show that $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

C. U. H. 89 ; R. U. H. 81.

Solution : We have $u = 3x^2y + 2x^2 - y^3 - 2y^2$.

Thus $\frac{\delta u}{\delta x} = 6xy + 4x = \phi_1(x, y) \quad \cdots (1)\;;\quad \frac{\delta^2 u}{\delta x^2} = 6y + 4 \quad \cdots (2)$

$\frac{\delta u}{\delta y} = 3x^2 - 3y^2 - 4y = \phi_2(x, y) \quad \cdots (3)\;;\quad \frac{\delta^2 u}{\delta y^2} = -6y - 4 \quad \cdots \cdots (4).$

Now $(2) + (4) \Rightarrow \frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0 \Rightarrow$ u is harmonic.

Now using Milne's method we have

$f'(z) = \phi_1(z, 0) - i\,\phi_2(z, 0)$ [Putting $x = z$, $y = 0$ in (1) and (3)]

$= 4z - 3iz^2$. Now integrating we have $f(z) = 2z^2 - iz^3 + ic$

$\Rightarrow f(z) = u + iv = 2(x + iy)^2 - i(x + iy)^3 + ic = 2(x^2 - y^2 + 2xyi)$

$- i(x^3 + 3x^2yi - 3xy^2 - iy^3) + ic \Rightarrow v = 4xy - x^3 + 3xy^2 + c.$

Example 118 : Show that the function $u = 2x - x^3 + 3xy^2$ is harmonic and also find the harmonic conjugate v if $f(z) = u + iv$ is analytic.

D. U. H. 91 ; J. U. H. 87, 89 ; C. U. H. 85.

Solution : We have $\frac{\delta u}{\delta x} = 2 - 3x^2 + 3y^2 = \phi_1(x, y)$ (say),

$\frac{\delta^2 u}{\delta x^2} = -6x$, $\frac{\delta u}{\delta y} = 6xy = \phi_2(x, y)$ (say) and $\frac{\delta^2 \psi}{\delta y^2} = 6x$.

Thus $\frac{\delta^2 \phi}{\delta x^2} + \frac{\delta^2 \phi}{\delta y^2} = 0 \Rightarrow$ u is harmonic. By Milne method, we have $f'(z) = \phi_1(z, 0) - i\,\phi_2(z, 0) = 2 - 3z^2 \Rightarrow f(z) = u + iv$

$= 2z - z^3 + c = 2(x + iy) - (x + iy)^3 + ic = 2(x + iy) - (x^3 + 3x^2yi$

$- 3xy^2 - iy^3) + ic \Rightarrow v = 2y - 3x^2y + y^3 + c$ where c is a constant.

Example 119 : Show that $u = \frac{y}{x^2 + y^2}$ is harmonic and find its harmonic conjugate v and $f(z) = u + iv$ if $f(z)$ is analytic.

D. U. H. T. 76.

Solution : We have $\dfrac{\delta u}{\delta x}=\dfrac{-2xy}{(x^2+y^2)^2}=\phi_1(x, y)$ (say),

$\dfrac{\delta^2 u}{\delta x^2}=\dfrac{2y(3x^2-y^2)}{(x^2+y^2)^3}$, $\dfrac{\delta u}{\delta y}=\dfrac{x^2-y^2}{(x^2+y)^2}=\phi_2(x, y)$ (say), $\dfrac{\delta^2 u}{\delta y^2}$

$=\dfrac{2y(y^2-3x^2)}{(x^2+y^2)^3}$. Then $\dfrac{\delta^2 u}{\delta x^2}+\dfrac{\delta^2 u}{\delta y^2}=0 \Rightarrow u$ is harmonic.

Now by Milne method, we have, $f'(z)=\phi_1(z, 0)-i\,\phi_2(z, 0)$

$=\dfrac{-i}{z^2} \Rightarrow f(z)=\dfrac{i}{z}+c=u+iv=\dfrac{i}{x+iy}+c_1+ic_2=\dfrac{i(x-iy)}{x^2+y^2}+c_1$

$+ic_2 \Rightarrow v=\dfrac{x}{x^2+y^2}+c_2.$

Example 120 : Find the harmonic conjugate of the function

$u=e^{x^2-y^2}\cos 2xy$ and the corresponding analytic function

$f(z)=u+i\,v$. ***J. U. H. 87 ; D. U. H. T. 78.***

Solution : We have $\dfrac{\delta u}{\delta x}=2e^{x^2-y^2}(x\cos 2xy-y\sin 2xy)$

$=\phi_1(x, y)$ (say), $\dfrac{\delta^2 u}{\delta x^2}=2e^{x^2-y^2}\,[2(x^2-y^2)\cos 2xy-4xy\sin 2xy$

$+\cos 2xy]$, $\dfrac{\delta u}{\delta y}=-2e^{x^2-y^2}(y\cos 2xy+x\sin 2xy)=\phi_2(x, y)$ (say),

$\dfrac{\delta^2 u}{\delta x^2}=2e^{x^2-y^2}\,[2(x^2-y^2)\cos 2xy+\cos 2xy-4xy\sin 2xy]$

and $\dfrac{\delta^2 u}{\delta y^2}=-2e^{x^2-y^2}\,[2(x^2-y^2)\cos 2xy+\cos 2xy-4xy\sin 2xy]$.

Now $\dfrac{\delta^2 u}{\delta x^2}+\dfrac{\delta^2 u}{\delta y^2}=0 \Rightarrow u$ is harmonic. Thus the harmonic conjugate v of u exists

Now by Milne method, we have $f'(z)=\phi_1(z,0)-i\phi_2(z,0)$ $=2ze^{z^2}=e^{z^2}d(z^2)\Rightarrow f(z)=u+iv=e^{z^2}+c=e^{x^2-y^2+2ixy}$ $+c=e^{x^2-y^2}[\cos 2xy+i\sin 2xy+c_1+ic_2 \Rightarrow v=e^{x^2-y^2}\sin 2xy$ $+c_2$ where $c=c_1+ic_2$.

Example 121 : Show that the function $u=x^2-y^2-2xy-2x+3y$ is harmonic and find the harmonic conjugate v also $f(z)=u+iv$ if $f(z)$ is analytic.

D. U. M. SC. P. T. 91.

Solution : We have $\frac{\delta u}{\delta x}=2x-2y-2=\phi_1(x,y)$ (say),

$$\frac{\delta^2 u}{\delta x^2}=-2,\ \frac{\delta u}{\delta y}=-2y-2x+3=\phi_2(x,y)\ (\text{say}),\ \frac{\delta^2 u}{\delta y^2}=-2,$$

Thus $\frac{\delta^2 u}{\delta x^2}+\frac{\delta^2 u}{\delta y^2}=0\Rightarrow u$ is harmonic.

Now by Milne method, we have $f'(z)=\phi_1(z,0)-i\,\phi_2(z,0)$ $=(2z-2)-i(-2z+3)=2(1+i)z-(2+3i)\Rightarrow f(z)=u+iv$ $=(1+i)z^2-(2+3i)z+c=(1+i)(x^2-y^2+2xyi)-(2+3i)(x+iy)$ $+c_1+ic_2\Rightarrow v=x^2-y^2+2xy-3x-2y+c_2$ where $c=c_1+ic_2$.

Example 122 : Show that the function $u=e^x(x\cos y-y\sin y)$ is a harmonic function and find the corresponding analytic function $f(z)=u+iv$. From it find v. ✓

C. U. H. 85, 88 ; D. U. H. T. 88 ; J. U. M. 91.

Soluiton : We have $u=e^x(x\cos y-y\sin y)$.

Then $\frac{\delta u}{\delta x}=e^x(x\cos y-y\sin y+\cos y)=\phi_1(x,y)$ ... (1)

$\frac{\delta^2 u}{\delta x^2}=e^x(x\cos y-y\sin y+2\cos y)$ ... (2)

$$\frac{\delta u}{\delta y}=e^x(-x\sin y-\sin y-y\cos y)=\phi_2(x,y) \quad \cdots \quad (3)$$

$$\frac{\delta^2 u}{\delta y^2}=e^x(-x\cos y+y\sin y-2\cos y) \quad \cdots \quad \cdots \quad (4)$$

Now $(2)+(4) \Rightarrow \frac{\delta^2 u}{\delta x^2}+\frac{\delta^2 u}{\delta y^2}=0 \Rightarrow u$ is a harmonic function.

Now using ***Milne's*** method, we have $f'(z)=\phi_1(z,0)-i\phi_2(z,0)$ $\left[\text{putting } x=z,\ y=0 \text{ in } (1) \text{ and } (3)\right]$

$=e^z(z+1)$. Now integrating we have $f(z)=e^z(z+1)$ $-e^z+c=ze^z+c$. Again we have, $f(z)=u+iv$

$=(x+iy)\,e^{x+iy}+c=(x+iy)e^x(\cos y+i\sin y)+c_1+ic_2 \Rightarrow$ $v=e^x(x\sin y+y\cos y)+c_2$ where $c=c_1+ic_2$.

Example 123 : Show that the function $u=2x(1-y)$ is harmonic and find a function v such that $f(z)=u+iv$ is analytic. Also find $f(z)$ in terms of z. ***D. U, H. 83 ; D. U. S. 83.***

Solution : We have $u=2x(1-y)$. Then $\frac{\delta u}{\delta x}=2(1-y)$

$=\phi_1(x,y)$ (say) $\cdots$ (1), $\frac{\delta^2 u}{\delta x^2}=0$ $\cdots$ (2),

$\frac{\delta u}{\delta y}=-2x=\phi_2(x,y)$ (say) $\cdots$ (2), $\frac{\delta^2 u}{\delta y^2}=0$ $\cdots$ $\cdots$ (4)

Now $(2)+(4) \Rightarrow \frac{\delta^2 u}{\delta x^2}+\frac{\delta^2 u}{\delta y^2}=0 \Rightarrow u$ is harmonic.

Now putting $x=z$ and $y=0$ in (1) and (2) $\Rightarrow \phi_1(z,0)=2$ and $\phi_2(z,0)=-2z$. Then $f^1(z)=\phi_1(z,0)-i\phi_2(z,0)=2+2iz$. Now integrating we get $f(z)=2z+iz^2+c \Rightarrow f(z)=u+iv=2(x+iy)+$ $i(x^2-y^2+2xyi)+c \Rightarrow v=x^2-y^2+2y+c_2$ where $c=c_1+ic_2$ (say).

Example 124 : Show that $u=e^{-x}(x\sin y-y\cos y)$ is harmonic. Find v such that $f(z)=u+iv$ is analytic.
D. U. H. T. 89 ; D. U. 81, 82 ; C. U. 82 ; R. U. M. SC. P. 85 ; D. U, M. SC. P. T. 90.

Final

Solution : We have $u = e^{-x}(x \sin y - y \cos y)$.

Then $\dfrac{\delta u}{\delta x} = e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y = \phi_1(x, y)$ ... (1)

$\dfrac{\delta^2 u}{\delta x^2} = -2e^{-x} \sin y + xe^{-x} \sin y - ye^{-x} \cos y$ ... ... (2),

$\dfrac{\delta u}{\delta y} = xe^{-x} \cos y + ye^{-x} \sin y - e^{-x} \cos y = \phi_2(x, y)$ ... (3),

$\dfrac{\delta^2 u}{\delta y^2} = -xe^{-x} \sin y + 2e^{-x} \sin y + e^{-x} \cos y$ ... ... (4)

Now (2)+(4) $\Rightarrow \dfrac{\delta^2 u}{\delta x^2} + \dfrac{\delta^2 u}{\delta y^2} = 0 \Rightarrow u$ is harmonic.

Now putting $x = z$ and $y = 0$ in (1) and (3), we get $\phi_1(z, 0) = 0$ and $\phi_2(z, 0) = (z-1)e^{-z}$.

Then $f'(z) = \phi_1(z, 0) - i\,\phi_2(z, 0) = -i(z-1)e^{-z}$.

Now integrating we get $f(z) = i(z-1)e^{-z} + i\,e^{-z} + c = ize^{-z} + c$.

Again we have $f(z) = u + iv = i(x+iy)e^{-(x+iy)} + c$

$= i(x+iy)e^{-x}(\cos y - i \sin y) + c \Rightarrow v = x \cos y\, e^{-x} + y \sin y\, e^{-x} + c_2$

$= e^{-x}(x \cos y + y \sin y) + c_2$ where $c = c_1 + i\,c_2$.

Here $z = 3$ is a simple zero and $z = -5$ is a zero of order 2.

179. Singular point or critical point or singularity of a function :

D. U. H. T. 89; D. U. H. 90.

A point at which a function $f(z)$ fails or ceases to be analytic is called a singular point.

Or equivalently, a point $z = z_0$ is called a singular point of a function $f(z)$, if $f(z)$ is not differentiable at z_0 but if every neighbourhood of z_0 contains any other point at which $f(z)$ must be differentiable.

Example 145 : Let $f(z) = \frac{z^2 - 4}{(z - 1)(z - 3)}$.

Here $z = 1$ and $z = 2$ are two singular points since at these two points the function are not analytic.

180. Various types of singularities :

(i) Isolated singular point or singularity :

A singular point $z = z_0$ is called an isolated singular point of a function $f(z)$ if we can find $\delta > 0$ such that the circle $|z - z_0| = \delta$ encloses no other singularity of the function other than z_0. If there exists no such δ, in this case z_0 is called a **non - isolated singularity** of $f(z)$.

Example 146 : Let $f(z) = \frac{1}{(z - 1)(z - 3)}$, where $z = 1$ and $z = 3$ are the singular points of $f(z)$. Here $z = 1$ is an isolated singular point of $f(z)$ since there exists at least one circle $|z - 1| = \delta < 2$ which does not include the singularity $z = 3$. Similarly, the circle $|z - 3| = \delta' < 2$ does not include the singularity $z = 1$ and therefore $z = 3$ is also an isolated singularity of $f(z)$.

Ordinary point : A point z_0 is called an ordinary point of a function $f(z)$ if z_0 is not a singular point and we can find $\delta > 0$ such that the circle $|z - z_0| = \delta$ encloses no singular point.

(ii) Pole : D. U. H. 83; D. U. H. T. 89; R. U. H. 76, 77.

A singular point $z = z_0$ is called a pole of order n of the function $f(z)$ if $\lim\limits_{z \to z_0} (z - z_0)^n f(z) = A \neq 0$, where n is a positive integer. If $n = 1$, then $z = z_0 =$ is called a **simple pole.** If $n = 2$, then $z = z_0$ is called a **double pole** or **pole of order 2.** Again, if $n = 3$, then $z = z_0$ is called a **triple pole** or **pole of order 3** and similarly we can define the poles of order 4, 5, 6, etc. If a is a zero of the function $f(z)$ of order n, then it is a pole of the function $1/f(z)$ of order n.

Example 147 : Let $f(z) = \dfrac{z - 3}{(z - 7)^5}$. Here $z = 7$ is a pole of order 5 of the function $f(z)$ since $\lim\limits_{z \to 7} (z - 7)^5 f(z)$

$$= \lim_{z \to 7} (z - 3) = 4 \neq 0.$$

Example 148 : Let $f(z) = \dfrac{z^2 + 5}{(z - 1)(z - 3)^4 (z - 4)}$. Here $z = 1$ and $z = 4$ are the simple poles of the function $f(z)$ and $z = 3$ is a pole of order 4 of $f(z)$.

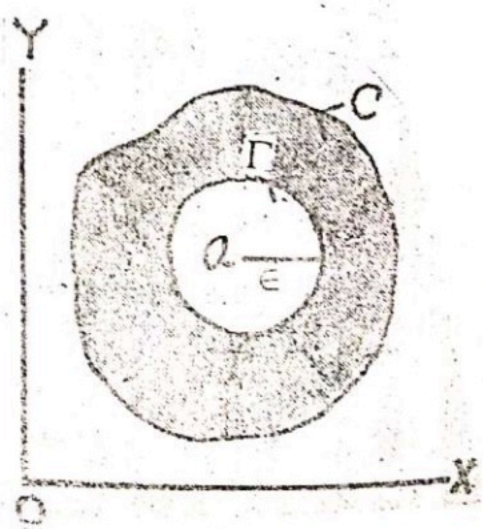
CHPTER-5

CAUCHY'S INTEGRAL FORMULAE AND RELATED THEOREMS

Cauchy's integral formula (Theorem–123) : Let $f(z)$ be analytic inside and on a simple closed curve C. If a is any point inside C, then $f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a}\, dz$, where C is traversed in the positive sense, i. e. in the counter clockwise direction.

D. U. H. T. 75, 83, 85,88, 91; D. U. H. 86, 88, 89; J. U. H. 83, 85, 85, 87, 88; R. U. H. 73, 75, 77; C. U. H. 81, 86, 87; D. U. 72; R. U. M. Sc. P. 84, 86; R. U. 80, 82.

First proof : Here the function $\frac{f(z)}{z-a}$ is analytic inside and on the simple closed curve C except at the point z = a. Then by a corollary of Cauchy's fundamental integral theorem, we have $\oint_C \frac{f(z)}{z-a}\, dz = \oint_\Gamma \frac{f(z)}{z-a}\, dz \cdots (1)$, where Γ is



a circle of radius ε with centre at $z = a$ inside C. Now the equation of Γ is $|z-a| = \varepsilon \Rightarrow z - a = \varepsilon e^{i\theta} \Rightarrow z = a + \varepsilon e^{i\theta}$

where $0 \le \theta \le 2\pi$. Now putting $z = a + \varepsilon e^{i\theta}$, $dz = i\varepsilon e^{i\theta} d\theta$ and limit of θ from o to 2π in (1), we have $\oint_{\Gamma} \frac{f(z)}{z-a} dz$

$= \int_0^{2\pi} \frac{f(a + \varepsilon e^{i\theta})}{\varepsilon e^{i\theta}} i\varepsilon e^{i\theta} d\theta = i \int_0^{2\pi} f(a + \varepsilon e^{i\theta}) d\theta$ ... (2). Now by (1)

and (2), we have $\oint_C \frac{f(z)}{z-a} dz = i \int_0^{2\pi} f(a + e^{i\theta}) d\theta$ ... (3). Now

taking the limit $\varepsilon \to 0$ of both sides of (3) and using the continuity of $f(z)$, we get $\oint_C \frac{f(z)}{z-a} = \lim_{\varepsilon \to 0} i \int_0^{2\pi} f(a + \varepsilon e^{i\theta}) d\theta$

$$= i \int_0^{2\pi} \lim_{\varepsilon \to 0} f(a + \varepsilon e^{i\theta}) d\theta = i \int_0^{2\pi} f(a) d\theta = 2\pi i f(a)$$

$\Rightarrow f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz$ and our required result is proved.

Second Proof : Since $f(z)$ is analytic inside C including the point a, then we have $f(z) = f(a) + (z - a) f'(a) + (z - a)\eta$ ... (1), where $\eta \to 0$ as $z \to a$.

Now $\frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz$

$$= \frac{1}{2\pi i} \oint_C \frac{f(a) + \{f(z) - f(a)\}}{z-a} dz$$

$$= \frac{1}{2\pi i} \oint_C \frac{f(a)}{z-a} dz + \frac{1}{2\pi i} \oint_C \frac{f(z) - f(a)}{z-a} dz$$

$$= \frac{f(a)}{2\pi i} \oint_C \frac{dz}{z-a} + \frac{1}{2\pi i} \oint_C \frac{(z-a) f'(a) + (z-a)\eta}{z-a} dz \text{ [by (1)]}$$

$$= \frac{f(a)}{2\pi i}(2\pi i) + \frac{f'(a)}{2\pi i}\oint_C dz + \frac{1}{2\pi i}\oint_C \eta\, dz \quad \left[\because \oint_C \frac{dz}{z-a} = 2\pi i\right]$$

$$= f(a) + 0 + \frac{1}{2\pi i}\oint_C \eta dz \quad \left[\begin{array}{l}\text{The second part is zero by}\\ \text{the Cauchy's funda-mental}\\ \text{integral theorem}\end{array}\right]$$

$$= f(a) + \frac{1}{2\pi i}\oint_C \eta dz.$$

Now we take C so small that for all points on it $|\eta| < \varepsilon$, then

$$\left|\frac{1}{2\pi i}\oint_C \frac{f(z)}{z-a}dz - f(a)\right| = \left|\frac{1}{2\pi i}\oint_C \eta dz\right| < \frac{\varepsilon}{2\pi}L \ldots (2),$$

where L is the length of C. In the equality of (2), the right-hand side tends to zero and the left-hand side is independent of ε. Therefore, it must be zero.

Thus $f(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-a}dz$.

Theorem -124 : If $f(z)$ is analytic inside and on the boundary C of a simply-connected closed region $\Re$, then

$$\frac{1}{2\pi i}\oint_C \left\{\frac{1}{z-a} \pm \frac{1}{z-b}\right\} f(z)\, dz$$

$$= \begin{cases} f(a) \pm f(b) & \text{if the points a and b are inside C} \\ f(a) & \text{if only the point a lies inside C} \\ \pm f(b) & \text{if only the point b lies inside C} \\ 0 & \text{if the points a and b are not inside C} \end{cases}$$

Proof : Try yourself.

Example 182 : Show that $\frac{1}{2\pi i}\oint_C \frac{e^z}{z-2}dz$.

Example 183 : Show that $\oint_C \frac{\sin 3z}{z+\pi/2}\,dz = 2\pi i$, where C is the circle $|z| = 5$.

R. U. H. '81.

Solution : Here $f(z) = \sin 3z$ is analytic inside and on the circle $|z| = 5$. Again $z = -\pi/2$ lies inside the given circle and therefore by Cauchy's integral formula, we have $\oint_C \frac{\sin 3z}{z+\pi/2}\,dz = 2\pi i\, f(-\pi/2) = 2\pi i \sin 3(-\pi/2) = 2\pi i$.

Example 184 : Show that $\oint_C \frac{e^{3z}}{z-\pi i}\,dz$ R. U. H. '81, '86.

$$\begin{cases} -2\pi i & \text{if C is the circle } |z-1| = 4 \\ 0 & \text{if C is the ellipse } |z-2| + |z+2| = 6. \end{cases}$$

Solution : (First part) : Let $f(z) = e^{3z}$ and it is analytic inside and on the given circle $|z-1| = 4$. Again $z = \pi i$ is a point inside the given circle. Then by the Cauchy's integral formula, we have $\oint_C \frac{e^{3z}}{z-\pi i} dz = 2\pi i \, f(\pi i) = 2\pi i \, e^{3\pi i} = -2\pi i$

since $e^{3\pi i} = \cos 3\pi + i\sin 3\pi = -1$.

Second part : Let $f(z) = \frac{e^{3z}}{z-\pi i}$. Here

$|z-2| + |z+2| = 6$ is an ellipse with foci at $(2, 0)$ and $(-2, 0)$ whose major axis has length 6. Therefore, the point $z = \pi i$ lies outside the ellipse. Thus $f(z)$ is analytic inside and on the ellipse C and by Cauchy's integral theorem $\oint_C f(z)\, dz = \oint_C \frac{e^{3z}}{z-\pi i} \cdot dz = 0$.

Example 185 : show that

$\oint_C \frac{\sin^6 z}{z-\pi/6} dz = \frac{\pi i}{32}$, where C is the circle $|z| = 1$.

Solution : Here $\sin^6 z$ is analytic inside and on the circle $|z| = 1$ and $z = \pi/6$ is a point inside the given circle. Then by the Cauchy's integral formula $\oint_C \frac{\sin^6 z}{z-\pi/6} dz =$ $2\pi i (\sin \pi/6)^6 = 2\pi i \left(\frac{1}{2}\right)^6 = \frac{\pi i}{32}$.

Example 186 : show that $\oint_C \frac{\cos z}{z-\pi} dz = -2\pi i$, where C is the circle $|z-1| = 3$.

Solution : Here $f(z) = \cos z$ and the point $z = a = \pi$ lies inside the given circle, then by the Cauchy's integral formula, we have $\frac{1}{2\pi i} \oint_C \frac{\cos z}{z-\pi} dz = \cos \pi = -1 \Rightarrow \oint_C \frac{\cos z}{z-\pi} dz$ $= -2\pi i$.

Example 188 : Show that $\oint_C \frac{e^z}{z(z+1)}\, dz = 2\pi i(1 - e^{-1})$,

where C is the circle $|z - 1| = 3$.

Solution : We have $\oint_C \frac{e^z}{z(z+1)}\, dz = \oint_C \left(\frac{1}{z} - \frac{1}{z+1}\right) e^z dz$

$$= \oint_C \frac{e^z}{z}\, dz - \oint_C \frac{e^z}{z+1}\, dz = 2\pi i e^0 - 2\pi i e^{-1} = 2\pi i(1 - e^{-1}),$$ by

the Cauchy's integral formula.

Cauchy's integral formula for the first derivative of an analytic function (Theorem — 125) : Let $f(z)$ be analytic inside and on a simple closed curve C and a is a point inside C. Then $f'(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^2}\,dz$.

R. U. 80; R. U. H. 73, '80, '82, '85; R. U. M. SC. P. '84, '85; D. U. M. SC. P. '89; D. U. M. SC. P. T. '88; D. U. H. T. '90; D. U. H. '87, '90; J. U. H. '91.

Proof : Let a be any point inside C and a + h be a neighbouring point of a, which is also inside C. Now by Cauchy's integral formula, we have $\frac{f(a+h) - f(a)}{h} =$

$$\frac{1}{2\pi i}\oint_C \frac{1}{h}\left\{\frac{1}{z-(a+h)} - \frac{1}{z-a}\right\} f(z)\,dz = \frac{1}{2\pi i}\oint_C \frac{f(z)dz}{(z-a)(z-a-h)}$$

and also $\frac{f(a+h) - f(a)}{h} - \frac{1}{2\pi i}\oint_C \frac{f(z)dz}{(z-a)^2}$

$$= \frac{h}{2\pi i}\oint_C \frac{f(z)dz}{(z-a)^2(a-a-h)} \quad \dots (1).$$

Now if $h \to 0$, then (1)

$$\Rightarrow \lim_{h\to 0} \frac{f(z+h) - f(a)}{h} - \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^2}\,dz = 0 \Rightarrow f'(a)$$

$= \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^2}\,dz$ provided that $\left|\oint_C \frac{f(z)\,dz}{(z-a)^2(z-a-h)}\right|$

is bounded as $h \to 0$. Since $f(z)$ is analytic, then it is continuous on C and it is bounded. Suppose that $|f(z)| \le M$ on C and that the shortest distance from a to C is δ, i. e. $|z - a| \ge \delta$. If L be the length of C and $|h| \le \delta$, then $\left| \oint_C \frac{f(z)}{(z-a)^2 (z-a-h)} dz \right| < \frac{ML}{\delta^2 (d - |h|)}$ which is bounded as $|h| \to 0$. Thus $f'(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)^2} dz$. Hence the theorem is proved.

Cauchy's integral formula for the second derivative of an analytic function (Theorem — 126) : Let $f(z)$ be analytic inside and on a simple closed curve C and a is a point inside C. Then $f''(a) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^3} dz$.

Proof : Let a be any point inside C and a + h be a neighbouring point of a, which is also inside C. Now by Cauchy's first derivative integral formula, we have

$$\frac{f'(a+h) - f'(a)}{h} = \frac{1}{2\pi i} \oint_C \frac{1}{h} \left\{ \frac{1}{(z-a-h)^2} - \frac{1}{(z-a)^2} \right\} f(z)$$

$$dz = \frac{2!}{2\pi i} \oint_C \frac{f(z)\, dz}{(z-a)^3} + \frac{h}{2\pi i} \oint_C \frac{3(z-a) - 2h}{(z-a-h)^2 (z-a)^3} f(z)\, dz \; \ldots (1)$$

Now if $h \to 0$, then (1) $\Rightarrow \lim\limits_{h \to 0} \frac{f'(a+h) - f'(a)}{h} - \frac{2!}{2\pi i} \oint_C \frac{f(z) dz}{(z-a)^3}$

$= 0 \Rightarrow f''(a) = \frac{2!}{2\pi i} \oint_C \frac{f(z) dz}{(z-a)^3}$ provided that

$\left|\int_C \frac{3(z-a)-2h}{(z-a-h)^2(z-a)^3} f(z)\,dz\right|$ is bounded as $h \to 0$. Since $f(z)$ is analytic, then it is continuous on C and it is bounded. Suppose $|\{3(z-a)-2h\} f(z)| \le M$ and that the shortest distance from a to C is δ, i. e. $|z-a| \ge \delta$. If L be the length of C and $|h| < \delta$, then $\left|\oint_C \frac{\{3(z-a)-2h\} f(z)dz}{(z-a-h)^2(z-a)^3}\right| \le \frac{ML}{(\delta - |h|)^2 \delta^3}$

which is bounded as $|h| \to 0$. Thus $f''(a)$

$= \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^3}\,dz$. Hence the theorem is proved.

Theorem -127 : Show that $f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)\,dz}{(z-a)^{n+1}}$ where $n = 0, 1, 2, 3, \ldots$.

D. U. H. '89, '90; J. U. H. '86, '88, '89, '90; R. U. H. '85; D. U. M. SC. P. '89.

Proof : BY the above three theorems show that the theorem is true for $n = 0, 1, 2$. In a similar way we can prove the theorem for $n = 3, 4, 5, \ldots$. The second proof is given below.

Cauchy's integral formula for the nth derivative of an analytic function (Theorem — 128) : Let $f(z)$ be analytic inside and on the boundary C of a simply-connected region $\mathfrak{R}$. Then $f(z)$ has, at every point $z = a$ of $\mathfrak{R}$, derivatives of all orders and their values are given by

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}}\,dz, \text{ where } n = 1, 2, 3, \ldots\ldots$$

J. U. H. '86, '88, '89, '90; R. U. H. '85; D. U. H. '89, '90; D. U. M. SC. P. '89.

Proof : Using mathematical induction, we will prove the theorem. If n = 1, then $f'(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^2}\,dz$ and the theorem is true for n = 1. Now we assume that the theorem is true for n = m. Then we have $f^{(m)}(a)$

$$= \frac{m!}{2\pi i}\oint_C \frac{f(z)}{(z-a)^{m+1}}\,dz. \text{ Now } \frac{f^{(m)}(a+h) - f^{(m)}(a)}{h}$$

$$= \frac{1}{h}\frac{m!}{2\pi i}\left(\oint_C \frac{f(z)}{(z-a-h)^{m+1}}\,dz - \oint_C \frac{f(z)}{(z-a)^{m+1}}\,dz\right)$$

$$= \frac{1}{h}\;\frac{m!}{2\pi i}\oint_C \left[\frac{1}{(z-a)^{m+1}}\right]\left\{\left(1-\frac{h}{z-a}\right)^{-(m+1)} - 1\right\} f(z)\,dz$$

$$= \frac{1}{h}\frac{m!}{2\pi i}\oint_C \left[\frac{1}{(z-a)^{m+1}}\left\{(1 + (m+1)\frac{h}{z-a} + \right.\right.$$

$$\left.\left.\frac{(m+1)(m+2)}{2!}\left(\frac{h}{z-a}\right)^2 + \ldots - 1\right\}\right] f(z)\,dz \ldots (1).$$

Now taking $h \to 0$ in both sides of (1), we have

$$\lim_{h\to 0} \frac{f^{(m)}(a+h) - f^{(m)}(a)}{h} = \frac{m!\,(m+1)}{2\pi i}\oint_C \frac{f(z)}{(z-a)^{m+2}}\,dz =$$

$$f^{(m+1)}(a) = \frac{(m+1)!}{2\pi i}\oint_C \frac{f(z)\,dz}{(z-a)^{m+2}} \ldots (2).$$

The above result (2) shows that the theorem is true for n = m + 1. But it is shown that it is true for n = 1 and hence it must be true for n = 1 + 1 = 2. But it is true for n = 2, then it is true for n = 2 + 1 = 3 and so on. Therefore, it is true for all integral values of n and we have

$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}} dz$. From the above result it is clear that if $f'(a)$ exists, then $f''(a)$, $f'''(a)$, ... , $f^{(n)}(a)$ are exist. Thus if a function of a complex variable has a first derivative (i. e. analytic) in a simply connected region $\Re$, then all its higher derivatives exist in $\Re$ (i. e. analytic in $\Re$). Hence the theorem is proved.

Example 196 : Show that $\frac{1}{2\pi i}\oint_C \frac{ze^{tz}}{(z+1)^3}\,dz = \left(t - \frac{t^2}{2}\right) e^{-t}$

where C is any simple closed curve enclosing $z = -1$ and $t > 0$. **R. U. H. '74; C. U. M. SC. P. '87.**

Solution : Here $f(z) = ze^{zt}$ is analytic inside and on C and $a = -1$ is any point inside C. Then using the Cauchy's integral formula $\frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^{n+1}}\,dz = \frac{1}{n!}\left[\frac{d^n}{dz^n} f(z)\right]$ at $z = a$

for $n = 2$, we have $\frac{1}{2\pi i}\oint_C \frac{ze^{tz}}{(z+1)^3}\,dz = \frac{1}{2!}\left[\frac{d^2}{dz^2}(ze^{tz})\right]$ at $z = -1 = \frac{1}{2}\left[\frac{d}{dz}(e^{tz} + tze^{tz})\right]$ at $z = -1 = \frac{1}{2}\left[2te^{tz} + t^2 ze^{tz}\right]$ at $z = -1 = \left[t - \frac{t^2}{2}\right] e^{-t}$.

Example 197 : Show that $\oint_C \frac{e^{2z}}{(z+1)^4}\,dz = \frac{8\pi i e^{-2}}{3}$, where C is the circle $|z| = 3$. **D. U. H. '91.**

Solution : Here $f(z) = e^{2z}$ is analytic inside and on the given circle and $z = a = -1$ is a point inside

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Example 199 : Show that $\oint_C \frac{\sin^6 z}{(z - \pi/6)^3} dz = \frac{21\pi i}{16}$ where C is the circle $|z| = 1$.

Solution : Here $f(z) = \sin^6 z$ is analytic inside and on the circle $|z| = 1$ and $z = a = \pi/6$ is a point inside the given circle. Then using the Cauchy's integral formula

$$\oint_C \frac{f(z)}{(z - a)^{n+1}} dz = \frac{2\pi i}{n!} \left[\frac{d^n}{dz^n} f(z)\right]_{\text{at } z = a} \text{ for } n = 2.$$

we have $\oint_C \frac{\sin^6 z}{(z-\pi/6)^3} dz = \frac{2\pi i}{2!}\left[\frac{d^2}{dz^2}\sin^6 z\right]_{\text{at} z = \pi/6}$

$= \pi i \left[\frac{d}{dz}(6\sin^5 z \cos z)\right]_{\text{at } z = \pi/6}$

$= \pi i \, [-6 \sin^6 z + 30 \sin^4 z \cos^2 z]$ at $z = \pi/6$

$= \pi i \left[-6\left(\frac{1}{2}\right)^6 + 30\left(\frac{1}{2}\right)^4\left(\frac{\sqrt{3}}{2}\right)^2\right] = \frac{(-6+90)\pi i}{64} = \frac{21\pi i}{16}.$

Example 200 : If $z = 1$ is a point inside a simple closed curve C, then show that $\oint_C \frac{5z^2 - 3z + 2}{(z-1)^3} dz = 10\pi i.$

Solution : Here $5z^2 - 3z + 2$ is analytic inside and on C. Then by the Cauchy's integral formula $\oint_C \frac{f(z)}{(z-a)^{n+1}} dz$

$= \frac{2\pi i}{n!}\left[\frac{d^n}{dz^n} f(z)\right]_{\text{at} z = a}$ for $n = 2$, we have $\oint_C \frac{5z^2 - 3z + 2}{(z-1)^3} dz$

$= \frac{2\pi i}{2!}\left[\frac{d^2}{dz^2}(5z^2 - 3z + 2)\right]_{\text{at } z = 1}$

$= \pi i \left[\frac{d}{dz}(10z - 3)\right]_{\text{at} z = 1} = \pi i \, [10] = 10\pi i.$

Ezample 201 : Show that $\left(\frac{a^n}{n!}\right)^2 = \frac{1}{2\pi i}\oint_C \frac{a^n e^{az}}{n!\, z^{n+1}}\, dz.$

C. U. H. '90; R. U. H. '73.

Solution : Let $f(z) = e^{az} \Rightarrow f^{(n)}(z) = a^n e^{az}$ ... (1).

Now putting $z = 0$ in (1) we get $a^n = f^{(n)}(0)$ ... (2)

But by Cauchy's integral formula,

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$f^{(n)}(b) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z-b)^{n+1}}\, dz \Rightarrow f^{(n)}(0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{z^{n+1}}\, dz$

... (3). Now by (2) and (3) we have $\frac{a^n}{n!} = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z^{n+1}}\, dz$ ... (4)

and multiplying both sides of (4) by $\frac{a^n}{n!}$ we get $\left(\frac{a^n}{n!}\right)^2$

$$= \frac{1}{2\pi i}\oint_C \frac{a^n e^{az}}{n!\, z^{n+1}}\, dz.$$

Hence our required result is proved.

Example 202 : Show that $\sum_{n=0}^{\infty}\left(\frac{a^n}{n!}\right)^2 = \frac{1}{2\pi}\int_0^{2\pi} e^{2a\cos\theta}\,d\theta.$

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C. U. H. '90.

Solution : We have $\left(\frac{a^n}{n!}\right)^2 = \frac{1}{2\pi i}\cdot\oint_C \frac{a^n e^{az}}{n!\,z^{n+1}}\,dz$ ... (1)

Now taking summation over n in (1) from o to ∞, then

we have $\sum_{n=0}^{\infty}\left(\frac{a^n}{n!}\right)^2 = \sum_{n=0}^{\infty}\frac{1}{2\pi i}\oint_C \frac{a^n e^{az}}{n!z^{n+1}}\,dz$

$$= \frac{1}{2\pi i}\oint_C e^{az}\sum_{n=0}^{\infty}\left(\frac{a}{z}\right)^n \frac{1}{n!}\frac{dz}{z} = \frac{1}{2\pi i}\oint_C e^{az}\,e^{a/z}\frac{dz}{z}$$

$$= \frac{1}{2\pi i}\oint_C e^{a\left(z+\frac{1}{z}\right)}\frac{dz}{z} \text{ ... (2).}$$

Now let C be a circle of unit radius having centre at the origin, then $|z| = 1 \Rightarrow z = e^{i\theta}$, $dz = ie^{i\theta}d\theta$, $z + \frac{1}{z} = e^{i\theta} + e^{-i\theta}$ $= 2\cos\theta$ and $0 \le \theta \le 2\pi$. Using these (2) $\Rightarrow \sum_{n=0}^{\infty}\left(\frac{a^n}{n!}\right)^2$

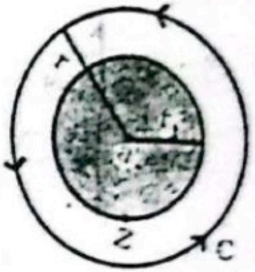
$= \frac{1}{2\pi i}\int_0^{2\pi} e^{2a\cos\theta}\frac{ie^{i\theta}\,d\theta}{e^{i\theta}} = \frac{1}{2\pi}\int_0^{2\pi} e^{2a\cos\theta}\,d\theta$. Hence the required result is proved.

Taylor's theorem (Theorem —190) : If $f(z)$ is analytic for all values of z inside a circle C with centre at a, then

$$f(z) = f(a) + (z-a)\, f'(a) + \frac{(z-a)^2}{2!} f''(a) + \frac{(z-a)^3}{3!} f'''(a) + \ldots$$

D. U. M. SC. P. '88; D. U. M. SC. P. T. '89; D. U. H. T. '75;
D. U. H '83; R. U. '76, 79; C. U. H. '81; R. U. H. '73, '82, '88;
R. U. M. SC. P. '84.

Proof : Let C be a circle of centre a and radius r. Let z be any point inside C such that $|z - a| = r_1 < r$. Then by the Cauchy's integral formula,



$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{w - z}\, dw =$$

$$\frac{1}{2\pi i} \oint_C \frac{f(w)}{(w-a) - (z-a)}\, dw$$

$$= \frac{1}{2\pi i} \oint_C \left\{1 - \frac{z-a}{w-a}\right\}^{-1} \frac{f(w)}{w-a}\, dw$$

$$= \frac{1}{2\pi i} \oint_C \Bigg[\Bigg\{1 + \frac{z-a}{w-a} + \left(\frac{z-a}{w-a}\right)^2 + \ldots$$

$$+ \left(\frac{z-a}{w-a}\right)^{n-1} + \left(\frac{z-a}{w-a}\right)^n \left\{1 - \frac{z-a}{w-a}\right\}^{-1}\Bigg] \frac{f(w)}{w-a}\, dw$$

$$= \frac{1}{2\pi i} \oint_C \Bigg[1 + \left(\frac{z-a}{w-a}\right) + \left(\frac{z-a}{w-a}\right)^2 + \ldots$$

$$+ \left(\frac{z-a}{w-a}\right)^{n-1} + \left(\frac{z-a}{w-a}\right)^n \frac{w-a}{w-z}\Bigg] \frac{f(w)}{w-a}\, dw$$

Q

Example-220 : Expand $f(z) = \log(1 + z)$ in a Taylor series about $z = 0$ and determine the region of convergence for the required series.

Solution : (First part) :

$f(z) = \log(1 + z)$. then $f(0) = 0$

$f'(z) = \dfrac{1}{1+z}$. $\therefore$ $f'(0) = 1$

$f''(z) = \dfrac{-1}{(1+z)^2}$. $\therefore$ $f''(0) = -1$

$f'''(z) = \dfrac{(-1)(-2)}{(1+z)^3}$. $\therefore$ $f'''(0) = (-1)(-2) = 2\,!$

$$f^{(n)}(z) = \frac{(-1)(-2)\ldots\{-(n-1)\}}{(1+z)^n}$$

$\therefore$

$$f^{(n)}(0) = (-1)(-2)\ldots\{-(n-1)\} = (-1)^{n-1}\,(n-1)\,!$$

Thus $f(z) = \log(1 + z)$

$$= f(0) + zf'(0) + \frac{z^2}{2\,!}f''(0) + \frac{z^3}{3\,!}f'''(0) + \ldots\ldots$$

$$= z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \ldots\ldots$$

Second part : Here n th term $= u_n = \dfrac{(-1)^{n-1}z^n}{n}$ and

$u_{n+1} = \dfrac{(-1)^n z^{n+1}}{n+1}$. Then we have $\dfrac{u_{n+1}}{u_n} = \dfrac{-nz}{n+1}$ and using the ratio test $\lim\limits_{n\to\infty} \left|\dfrac{u_{n+1}}{u_n}\right| = \lim\limits_{n\to\infty} \left|\dfrac{nz}{n+1}\right| = |z|$.

Thus the series is convergent for $|z| < 1$ and also it can be shown that the series is convergent for $|z| = 1$ except for $z = -1$. From this result it follows that the series converges in a circle which extends to the nearest singularity (i. e. $z = -1$) of $f(z)$.

Example-2.1 : Expand $\cos z$ about $z = \pi/2$ in a Taylor series. R. U. '76.

B

Solution : We have $f(z) = \cos z$. Then $f'(z) = -\sin z$, $f''(z) = -\cos z$, $f'''(z) = \sin z$, $f^{iv}(z) = \cos z$, ... and $f(\pi/2) = 0$, $f'(\pi/2) = -1$, $f''(\pi/2) = 0$, $f'''(\pi/2) = 1$, $f^{iv}(\pi/2) = 0$, ...

Then about $z = a = \pi/2$, we have

$$f(z) = f(a) + (z-a)\, f'(a) + \frac{(z-a)^2}{2!} f''(a) + \frac{(z-a)^3}{3!} f'''(a) + \dots\dots$$

$$\Rightarrow f(z) = -(z - \pi/2) + \frac{(z-\pi/2)^3}{3!} - \frac{(z-\pi/2)^5}{5!} + \dots\dots$$

$\Rightarrow f(z) = -(z-\pi/2) \quad 3! \quad 5!$

Example-222 : Expand sin z about $z = \pi/4$ in a Taylor series.

Solution : We have $f(z) = \sin z$. Then $f'(z) = \cos z$, $f''(z) = -\sin z$, $f'''(z) = -\cos z$, $f^{iv}(z) = \sin z$, ... and

$$f(\pi/4) = \frac{1}{\sqrt{2}},\ f'(\pi/4) = \frac{1}{\sqrt{2}},\ f''(\pi/4) = -\frac{1}{\sqrt{2}},$$

$$f'''(\pi/4) = -\frac{1}{\sqrt{2}} \cdots$$

Then about $z = a = \pi/4$, we have $f(z) =$

$$f(a) + (z-a)\,f'(a) + \frac{(z-a)^2}{2!}f''(a) + \frac{(z-a)^3}{3!}f'''(a) + \cdots\cdots$$

$$= \frac{1}{\sqrt{2}} + (z-\pi/4)\frac{1}{\sqrt{2}} - \frac{(z-\pi/4)^2}{2!}\frac{1}{\sqrt{2}} - \frac{(z-\pi/4)^3}{3!}\frac{1}{\sqrt{2}} + \cdots\cdots$$

$$= \frac{1}{\sqrt{2}}\left\{1 + (z-\pi/4) - \frac{(z-\pi/4)^2}{2!} - \frac{(z-\pi/4)^3}{3!} + \cdots\cdots\right\}$$

Example-223 : Expand $\log\left(\frac{1+z}{1-z}\right)$ in a Taylor series about $z = 0$. D. U. H. '88, '90. D. U. M. SC. P. T. '91.

Solution : If $|z| < 1$, then we have

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \frac{z^5}{5} - \cdots\cdots \text{(1) and}$$

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$$\log(1-z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \frac{z^4}{4} - \frac{z^5}{5} - \dots\dots \quad (2)$$

Now (1)–(2) $\Rightarrow \log(1+z) - \log(1-z) = 2\left(z + \frac{z^3}{3} + \frac{z^5}{5} + \dots\right)$

$\Rightarrow \log\left(\frac{1+z}{1-z}\right) = 2\left(z + \frac{z^3}{3} + \frac{z^5}{5} + \dots\dots\right)$ which converges for $|z| < 1$. It can be also proved that this series converge for $|z| = 1$ except for $z = -1$.

N. B. In the above example, in the case of multiple valued functions, the principal branch is used.

Laurent's theorem (Theorem —191) : If $f(z)$ is analytic inside and on the boundary of a ring-shaped region ℜ bounded by two concentric circles C_1 and C_2 of centre a and radii r_1 and r_2 $(r_2 < r_1)$ respectively' then for all z in ℜ,

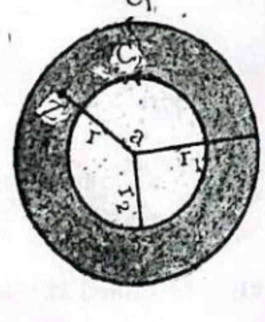
$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{a_{-n}}{(z-a)^n} \quad \text{where}$$

$$a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-a)^{n+1}}\, dw, \; n = 0, 1, 2, 3, \ldots$$

$$\text{and } a_{-n} = \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w-a)^{-n+1}}\, dw, \; n = 1, 2, 3, \ldots$$

D. U. H. '85, '88; R. U. M. SC. P. '85, '88; R. U. '79, 83; C. U. H. '82; D. U. '82; D. U. H. T. '82; R. U. H. '76.

Proof : Let the ring-shaped region ℜ = {|z − a| = r : $r_2 < r < r_1$ }. If z ε ℜ, then by the Cauchy's integral formula for multiply-connected region, i. e. for the ring-shaped region ℜ, we have $f(z) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w-z}\, dw$



$$- \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w-z}\, dw$$

$$= \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-a)-(z-a)}\, dw + \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(z-a)-(w-a)}\, dw$$

$$= \frac{1}{2\pi i} \oint_{C_1} \left\{1 - \frac{z-a}{w-a}\right\}^{-1} \frac{f(w)}{w-a}\, dw$$

$$+ \frac{1}{2\pi i} \oint_{C_2} \left\{1 - \frac{w-a}{z-a}\right\}^{-1} \frac{f(w)}{z-a}\, dw$$

$$= \frac{1}{2\pi i} \oint_{C_1} \left\{1 + \frac{z-a}{w-a} + \left(\frac{z-a}{w-a}\right)^2 + \ldots\right.$$

$$\left. + \left(\frac{z-a}{w-a}\right)^{n-1} + \left(\frac{z-a}{w-a}\right)^{n} \frac{w-a}{w-z}\right\} \frac{f(w)}{w-a}\, dw$$

$$+ \frac{1}{2\pi i} \oint_{C_2} \left\{1 + \frac{w-a}{z-a} + \left(\frac{w-a}{z-a}\right)^2 + \ldots\ldots\right.$$

$$\left. + \left(\frac{w-a}{z-a}\right)^{n-1} + \left(\frac{w-a}{z-a}\right)^{n} \frac{z-a}{z-w}\right\} \frac{f(w)}{z-a}$$

$$= \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w-a}\, dw + \frac{z-a}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-a)^2}\, dw + \ldots$$

$$+ \frac{(z-a)^{n-1}}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-a)^n}\, dw + U_n$$

$$+ \frac{1}{(z-a) 2\pi i} \oint_{C_2} f(w)\, dw + \frac{1}{(z-a)^2 2\pi i} \oint_{C_2} (w-a)\, f(w)\, dw + \ldots$$

$$+ \frac{1}{(z-a)^n\, 2\pi i} \oint_{C_2} (w-a)^{n-1} f(w)\, dw + V_n$$

$$= a_0 + a_1(z-a) + a_2 (z-a)^2 + \ldots + a_{n-1} (z-a)^{n-1} + U_n$$

$$+ \frac{a_1}{z-a} + \frac{a_2}{(z-a)^2} + \ldots + \frac{a_n}{(z-a)^n} + V_n \ldots (1)$$

where $a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-a)^{n+1}}\, dw$, n = 0, 1, 2, 3, ... ;

$$a_n = \frac{1}{2\pi i}\oint_{C_2} \frac{f(w)}{(w-a)^{-n+1}}\,dw,\ n = 1, 2, 3, \dots;$$

$$U_n = \frac{1}{2\pi i}\oint_{C_1} \left(\frac{z-a}{w-a}\right)^n \frac{f(w)}{w-z}\,dw \text{ and}$$

$$V_n = \frac{1}{2\pi i}\oint_{C_2} \left(\frac{w-a}{z-a}\right)^n \frac{f(w)}{z-w}\,dw.$$

Now $|w - z| = |(w-a) - (z-a)| \geq |w-a| - |z-a|$ for all points w on C_1. Then $|U_n|$ $\geq r_1 - r$ where $|w-a| = r_1$

$$= \frac{1}{2\pi}\left|(z-a)^n \oint_{C_1} \frac{f(w)}{(w-a)^n (w-z)}\,dw\right|$$

$$\leq \frac{r^2 M_1(2\pi r_1)}{2\pi r_1{}^n (r_1 - r)} = \frac{M_1}{1 - r/r_1}\left(\frac{r}{r_1}\right)^n \dots (2)$$ where M_1 is the maximum value of $f(w)$ on C_1 and $2\pi r_1$ is the length of the circle C_1. In(2) $\frac{r}{r_1} < 1$, then $\lim\limits_{n\to\infty} |U_n| \leq 0 \Rightarrow \lim\limits_{n\to\infty} U_n = 0 \dots (3)$.

Again $|z - w| = |(z-a) - (w-a)| \geq |z-a| - |w-a|$ for all points w on C. $\geq r - r_2$ where $|w-a| = r_2$

Then $$|V_n| = \frac{1}{2\pi}\left|\frac{1}{(z-a)^n}\oint_{C_2} (w-a)^n \frac{f(w)}{z-w}\,dw\right|$$

$$\leq \frac{r_2{}^n M_2(2\pi r_2)}{2\pi r^n (r - r_2)} = \frac{M_2}{(r/r_2) - 1}\left(\frac{r_2}{r}\right)^n \dots (4)$$

where M_2 is the maximum value of $f(w)$ on C_2 and $2\pi r_2$ is the length of the circle C_2. In (4) $\frac{r_2}{r} < 1$, then $\lim\limits_{n\to\infty} |V_n| \leq 0 \Rightarrow \lim\limits_{n\to\infty} V_n = 0 \dots (5)$ Now taking limit $n\to\infty$ in (1) and using (3) and (5) we have

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_n(z-a)^n + \dots$$
$$+ \frac{a_{-1}}{z-a} + \frac{a_{-2}}{(z-a)^2} + \dots + \frac{a_{-n}}{(z-a)^n} + \dots$$ and the theorem is proved.

Historical Note : Pierre Alphonse Laurrent (1813 – 1854) was a French mathematician.

Other forms of the Laurent's theorem :

(i). $f(z) = \sum\limits_{n=-\infty}^{\infty} a_n (z-a)^n$ where

$$a_n = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^{n+1}}\,dz,\ n = 0, \pm 1, \pm 2, \pm 3, \dots$$

(ii). Now writting $z = a + h$ in the Laurent series, we have

$$f(a+h) = \sum_{n=-\infty}^{\infty} a_n h^n = a_0 + a_1 h + a_2 h^2 + \dots + a_n h^n + \dots +$$
$$\frac{a_{-1}}{h} + \frac{a_{-2}}{h^2} + \dots + \frac{a_{-n}}{h^n} + \dots \dots$$

222. Laurent series : The series

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_n(z-a)^n + \dots + \frac{a_{-1}}{z-a}$$
$$+ \frac{a_{-2}}{(z-a)^2} + \dots + \frac{a_{-n}}{(z-a)^n} + \dots$$ is called the Laurent series or expansion with coefficients $a_n = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^{n+1}}\,dz$.

where $n = 0, \pm 1, \pm 2, \pm 3, \dots$

223. Analytic and principal part of the Laurent series : We have the Laurent series is $f(z) = \sum\limits_{n=0}^{\infty} a_n(z-a)^n +$

$\sum_{n=1}^{\infty} \frac{a_n}{(z-a)^n} = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_n(z-a)^n \dots$

$+ \frac{a_1}{z-a} + \frac{a_2}{(z-a)^2} + \dots + \frac{a_n}{(z-a)^n} + \dots$ (1)

Here the part $\sum_{n=0}^{\infty} a_n(z-a)^n = a_0 + a_1(z-a) + a_2(z-a)^2 +$ ... (2) is called the analytic part of the Laurent series. Again, the part $\sum_{n=1}^{\infty} \frac{a_n}{(z-a)^n} = \frac{a_1}{z-a} + \frac{a_2}{(z-a)^2} + \dots + \frac{a_n}{(z-a)^n} + \dots$ (3) is called the principal part of the Laurent series. If the principal part (3) is zero, then (1)

$\Rightarrow f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_n(z-a)^n + \dots$.

which is the Taylor's series. Thus if the principal part of the Laurent series is zero, then the Laurent series reduces to a Taylor series.

D. U. M. SC. P. '89; D. U. H. '82.

Example-225 : Expand the function $f(z) = \frac{1}{z-3}$ in a Laurent series for the region :

D. U. H. T. '83

(i). $|z| < 3$; (ii). $|z| > 3$.

Solution : (i) We have $|z| < 3 \Rightarrow 3 > |z|$ and using this condition, we have $f(z) = \frac{1}{z-3}$

$= -\frac{1}{3}\left(1 - \frac{z}{3}\right)^{-1} = -\frac{1}{3}\left(1 + \frac{z}{3} + \frac{z^2}{9} + \frac{z^3}{27} + \frac{z^4}{81} + \dots\right)$

$= -\frac{1}{3} - \frac{z}{9} - \frac{z^2}{27} - \frac{z^3}{81} - \dots\dots$

(ii). We have $|z| > 3$ and using this we have $f(z) = \frac{1}{z-3} =$

$\frac{1}{z}\left(1 - \frac{3}{z}\right)^{-1} = \frac{1}{z}\left(1 + \frac{3}{z} + \frac{9}{z^2} + \frac{27}{z^3} + \dots\right)$

$= \frac{1}{z} + \frac{3}{z^2} + \frac{9}{z^3} + \frac{27}{z^4} + \dots\dots$

Example 226 : Expand $f(z) = \frac{1}{z(z-2)}$ in a Laurent series for the region. (i). $0 < |z| < 2$; (ii). $|z| > 2$.

D. U. M. SC. P. T. '91.

Solution : Resolving into partial fractions, we have

$f(z) = \frac{1}{z(z-2)} = \frac{1}{2}\left[\frac{1}{z-2} - \frac{1}{z}\right]$ ...(1).

(i). We have $0 < |z| < 2$. Now using the condition $|z| < 2$,

(1) $\Rightarrow \frac{1}{z(z-2)} = \frac{1}{2}\left[-\frac{1}{2}(1 - z/2)^{-1} - \frac{1}{z}\right]$

$= \frac{1}{2}\left[-\frac{1}{2}\left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \frac{z^4}{16} + \dots\right) - \frac{1}{z}\right]$

$= -\frac{1}{2z} - \frac{1}{4} - \frac{z}{8} - \frac{z^2}{16} - \frac{z^3}{32} - \frac{z^4}{64} - \dots\dots$

(ii). We have $|z| > 2$ and using this condition (1) $\Rightarrow$

$\frac{1}{z(z-2)} = \frac{1}{2}\left[\frac{1}{z}\left(1 - \frac{2}{z}\right)^{-1} - \frac{1}{z}\right] = \frac{1}{2z}\left(1 - \frac{2}{z}\right)^{-1} - \frac{1}{2z}$

$= \frac{1}{2z}\left(1 + \frac{2}{z} + \frac{4}{z^2} + \frac{8}{z^3} + \frac{16}{z^4} + \dots\right) - \frac{1}{2z}$

$= \frac{1}{z^2} + \frac{2}{z^3} + \frac{4}{z^4} + \frac{8}{z^5} + \dots\dots$

উপরে কম এর, নিচে বেশি → Naturally
উপরে নিচে সমান → আসবে 1 আর.

Example 227 : Expand $f(z) = \frac{z}{(z+1)(z+2)}$ in a Laurent series for the region $1 < |z| < 2$.

D. U. H. T. '87.

Solution : We have $f(z) = \frac{z}{(z+1)(z+2)} = \frac{2}{z+2} - \frac{1}{z+1}$ ... (1). But we have $1 < |z| < 2 \Rightarrow 2 > |z|$ and $|z| > 1$ ... (2). Then using the condition (2), (1) $\Rightarrow$

Final

$$f(z) = \frac{2}{2}\left(1+\frac{z}{2}\right)^{-1} - \frac{1}{z}\left(1+\frac{1}{z}\right)^{-1}$$

$$= \left(1-\frac{z}{2}+\frac{z^2}{2^2}-\frac{z^3}{2^3}+\dots\right) - \frac{1}{z}\left(1-\frac{1}{z}+\frac{1}{z^2}-\frac{1}{z^3}+\dots\right)$$

$$= \left(1-\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}-\dots\right) - \left(\frac{1}{z}-\frac{1}{z^2}+\frac{1}{z^3}-\frac{1}{z^4}+\dots\right)$$

$$= \dots\dots-\frac{1}{z^3}+\frac{1}{z^2}-\frac{1}{z}+1-\frac{z}{2}+\frac{z^2}{4}-\frac{z^3}{8}+\dots\dots$$

Example 228 : Expand the function $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series for the following given region : (i). $1<|z|<3$; (ii). $|z|>3$; (iii). $0<|z+1|<2$; (iv). $|z|<1$.

R. U. '83; R. U. M. SC. P. '84; D. U. H. '86, '90; C. U. H. '89.

Solution : Resolving into partial fractions, we have

$$\frac{1}{(z+1)(z+3)} = \frac{\frac{1}{-1+3}}{z+1} + \frac{\frac{1}{-3+1}}{z+3} = \frac{1}{2}\left(\frac{1}{z+1}\right) - \frac{1}{2}\left(\frac{1}{z+3}\right)$$

... (1).

(i). We have $1<|z|<3 \Rightarrow |z|>1$ ... (2) and $3>|z|$ ... (3). In the first term of the right hand side of (1), we will use the condition (2) and in the second we will use the condition (3). Now using (2) and (3), the equation (1) $\Rightarrow$

$$\frac{1}{(z+1)(z+3)} = \frac{1}{2z}\left(1+\frac{1}{z}\right)^{-1} - \frac{1}{6}\left(1+\frac{z}{3}\right)^{-1}$$

$$= \frac{1}{2z}\left(1-\frac{1}{z}+\frac{1}{z^2}-\frac{1}{z^3}+\dots\right) - \frac{1}{6}\left(1-\frac{z}{3}+\frac{z^2}{3^2}-\frac{z^3}{3^3}+\dots\right)$$

$$= \left(\frac{1}{2z}-\frac{1}{2z^2}+\frac{1}{2z^3}-\frac{1}{2z^4}+\dots\right) - \left(\frac{1}{6}-\frac{z}{18}+\frac{z^2}{54}-\frac{z^3}{162}+\dots\right)$$

(ii). We have $|z|>3$ ... (4). Both in the first and second term of the right hand side of (1), we will use the condition (4) and using this, we have $\frac{1}{(z+1)(z+3)}$

$$= \frac{1}{2z}\left(1+\frac{1}{z}\right)^{-1} - \frac{1}{2z}\left(1+\frac{3}{z}\right)^{-1} = \frac{1}{2z}\left(1-\frac{1}{z}+\frac{1}{z^2}-\frac{1}{z^3}+\dots\right)$$

$$-\frac{1}{2z}\left(1-\frac{3}{z}+\frac{9}{z^2}-\frac{27}{z^3}+\dots\right) = \left(\frac{1}{2z}-\frac{1}{2z^2}+\frac{1}{2z^3}-\frac{1}{2z^4}+\dots\right)$$

$$-\left(\frac{1}{2z}-\frac{3}{2z^2}+\frac{9}{2z^3}-\frac{27}{2z^4}+\dots\right) = \left(\frac{1}{2z}-\frac{1}{2z}\right) + \left(\frac{3}{2}-\frac{1}{2}\right)\frac{1}{z^2}$$

$$-\left(\frac{9}{2}-\frac{1}{2}\right)\frac{1}{z^3} + \left(\frac{27}{2}-\frac{1}{2}\right)\frac{1}{z^4} - \dots = \frac{1}{z^2}-\frac{4}{z^3}+\frac{13}{z^4}-\frac{40}{z^5}+\dots$$

(iii). We have $0<|z+1|<2 \Rightarrow 2>|z+1|$. Now using this condition (1) $\Rightarrow \frac{1}{(z+1)(z+3)}$

$$= \frac{1}{2(z+1)} - \frac{1}{2}\left\{\frac{1}{(z+1)+2}\right\}$$

$$= \frac{1}{2(z+1)} - \frac{1}{4}\left\{1+\left(\frac{z+1}{2}\right)\right\}^{-1} = \frac{1}{2(z+1)}$$

$$-\frac{1}{4}\left\{1-\frac{1}{2}(z+1)+\frac{1}{4}(z+1)^2-\frac{1}{8}(z+1)^3+\dots\right\}$$

$$= \frac{1}{2(z+1)} - \frac{1}{4} + \frac{1}{8}(z+1) - \frac{1}{16}(z+1)^2 + \dots$$

(iv). We have $|z|<1 \Rightarrow |z|<1<3$ and using these (1)

$$\Rightarrow \frac{1}{(z+1)(z+3)} = \frac{1}{2}(1+z)^{-1} - \frac{1}{6}(1+z/3)^{-1}$$

$$= \frac{1}{2}(1-z+z^2-z^3+\dots) - \frac{1}{6}\left(1-\frac{z}{3}+\frac{z^2}{9}-\frac{z^3}{27}+\dots\right)$$

$$= \left(\frac{1}{2}-\frac{1}{6}\right) - \left(\frac{1}{2}-\frac{1}{18}\right)z + \left(\frac{1}{2}-\frac{1}{54}\right)z^2 - \left(\frac{1}{2}-\frac{1}{162}\right)z^3 + \dots$$

$$= \frac{1}{3} - \frac{4}{9}z + \frac{13}{27}z^2 - \frac{40}{81}z^3 + \dots$$

Example-229 : Expand $f(z) = \frac{1}{z^2(z-1)(z+2)}$ in a Laurent series in the region $|z| > 2$. **D. U. H. T. '78.**

Solution : We have $\frac{1}{z^2(z-1)(z+2)} = \frac{1}{z^2}\left[\frac{1}{3(z-1)} - \frac{1}{3(z+2)}\right]$

$$= \frac{1}{3z^2}\left[\frac{1}{z}\left(1-\frac{1}{z}\right)^{-1} - \frac{1}{z}\left(1+\frac{2}{z}\right)^{-1}\right] \quad [\because |z| > 2]$$

$$= \frac{1}{3z^3}\left[\left(1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}+\dots\right) - \left(1-\frac{2}{z}+\frac{4}{z^2}-\frac{8}{z^3}+\dots\right)\right]$$

$$= \frac{1}{3z^3}\left(\frac{3}{z} - \frac{3}{z^2} + \frac{9}{z^3} - \frac{15}{z^4} + \frac{33}{z^5}\dots\right)$$

$$= \frac{1}{z^4} - \frac{1}{z^5} + \frac{3}{z^6} - \frac{5}{z^7} + \frac{11}{z^8} - \dots$$

Example-230 : Expand $f(z) = \frac{z^2}{(z-1)(z-2)}$ in a Laurent series for the region $1 < |z| < 2$. **D. U. H. T. '71, '85, '88.**

Solution : We have $f(z) = \frac{z^2}{(z-1)(z-2)} = 1 - \frac{1}{z-1} + \frac{4}{z-2}$ ... (1). But we have $1 < |z| < 2 \Rightarrow |z| > 1$ and $2 > |z|$ ... (2).

Now using the condition (2), (1) $\Rightarrow f(z) = \frac{z^2}{(z-1)(z-2)}$

$$= 1 - \frac{1}{z}\left(1-\frac{1}{z}\right)^{-1} - \frac{4}{2}\left(1-\frac{z}{2}\right)^{-1}$$

$$= 1 - \frac{1}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}+\dots\right) - 2\left(1+\frac{z}{2}+\frac{z^2}{2^2}+\frac{z^3}{2^3}+\dots\right)$$

$$= \left(1-\frac{1}{z}-\frac{1}{z^2}-\frac{1}{z^3}-\frac{1}{z^4}-\dots\right) - \left(2+z+\frac{z^2}{2}+\frac{z^3}{2^2}+\dots\right)$$

$$= \dots\dots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - 1 - z - \frac{z^2}{2} - \frac{z^3}{2^2} - \dots\dots$$

Example-231 : Expand $f(z) = \frac{z^3}{(z+1)(z-2)}$ in a Laurent series in the powers of $(z+1)$ in the region $0 < |z+1| < 3$. **D. U. H. T. '76.**

Solution : We have $f(z) = \frac{z^3}{(z+1)(z-2)} = \frac{(z^3-8)+8}{(z+1)(z-2)}$

$$= \frac{(z-2)(z^2+2z+4)+8}{(z+1)(z-2)} = \frac{z^2+2z+4}{z+1} + \boxed{\frac{8}{(z+1)(z-2)}}$$

$$= \frac{(z+1)^2+3}{z+1} - \frac{8}{3(z+1)} + \frac{8}{3(z-2)} \quad [\text{ by cover up rule }]$$

$$= (z+1) + \frac{3}{z+1} - \frac{8}{3(z+1)} + \frac{8}{3}\left\{\frac{1}{(z+1)-3}\right\}$$

$$= (z+1) + \frac{1}{3(z+1)} - \frac{8}{9}\left\{1-\left(\frac{z+1}{3}\right)\right\}^{-1} \quad [\because 3 > |z+1|]$$

$$= (z+1) + \frac{1}{3(z+1)} - \frac{8}{9}\left\{1+\frac{z+1}{3}+\frac{(z+1)^2}{9}+\frac{(z+1)^3}{27}+\dots\right\}$$

$$= \frac{1}{3(z+1)} - \frac{8}{9} + \frac{19}{27}(z+1) - \frac{8}{9}\left\{\frac{(z+1)^2}{9}+\frac{(z+1)^3}{27}+\dots\dots\right\}$$

Example-232 : Expand $f(z) = \frac{1}{(z^2+1)(z+2)}$ in a Laurent series for the region $1 < |z| < 2$. **D. U. H. T. '78.**

Solution : We have $f(z) = \frac{1}{(z^2+1)(z+2)} = \frac{1}{5}\left[\frac{1}{z+2} - \frac{z-2}{z^2+1}\right]$

$$= \frac{1}{5}\left[\frac{1}{2}\left(1+\frac{z}{2}\right)^{-1} - \frac{z-2}{z^2}\left(1+\frac{1}{z^2}\right)^{-1}\right]$$

$$[\because 2 > |z| \text{ and } |z| > 1]$$

$$= \frac{1}{10}\left(1+\frac{z}{2}\right)^{-1} - \frac{z-2}{5z^2}\left(1+\frac{1}{z^2}\right)^{-1}$$

$$= \frac{1}{10}\left(1-\frac{z}{2}+\frac{z^2}{2^2}-\frac{z^3}{2^3}+\dots\right) - \frac{z-2}{5z^2}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\dots\right).$$

Example 233 : Expand the function $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ for the following regions :

(i). $2 < |z| < 3$; (ii). $|z| < 2$; (iii). $|z| > 3$.

D. U. M. SC. P. T. '89; D. U. M. SC. P. '88; R. U. M. SC. P. '84; R. U. H. '80, '85; D. U. H. '90.

Solution : We have $f(z) = \frac{z^2-1}{(z+2)(z+3)} = 1 + \frac{3}{z+2} - \frac{8}{z+3}$ ... (1)

(i). We have $2 < |z| < 3 \Rightarrow |z| > 2$ and $3 > |z|$ ... (2).

Now using (2), (1) $\Rightarrow f(z) = 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1}$

$$= 1 + \frac{3}{z}\left(1-\frac{2}{z}+\frac{2^2}{z^2}-\frac{2^3}{z^3}+\dots\right) - \frac{8}{3}\left(1-\frac{z}{3}+\frac{z^2}{3^2}-\frac{z^3}{3^3}+\dots\right),$$

$$= \dots - \frac{24}{z^4} + \frac{12}{z^3} - \frac{6}{z^2} + \frac{3}{z} - \frac{5}{3} + \frac{8z}{3^2} - \frac{8z^2}{3^3} + \frac{8z^3}{3^4} - \dots$$

(ii). We have $|z| < 2$. Now using this (1) $\Rightarrow$

$$f(z) = 1 + \frac{3}{2}\left(1+\frac{z}{2}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1} \quad [\because |z| < 2 < 3]$$

$$= 1 + \frac{3}{2}\left(1-\frac{z}{2}+\frac{z^2}{2^2}-\frac{z^3}{2^3}+\dots\right) - \frac{8}{3}\left(1-\frac{z}{3}+\frac{z^2}{3^2}-\frac{z^3}{3^3}+\dots\right)$$

$$= \left(1+\frac{3}{2}-\frac{8}{3}\right) - \left(\frac{3}{2^2}-\frac{8}{3^2}\right)z + \left(\frac{3}{2^3}-\frac{8}{3^3}\right)z^2 - \left(\frac{3}{2^4}-\frac{8}{3^4}\right)z^3 + \dots$$

$$= -\frac{1}{6} - \left(\frac{3}{2^2}-\frac{8}{3^2}\right)z + \left(\frac{3}{2^3}-\frac{8}{3^3}\right)z^2 - \left(\frac{3}{2^4}-\frac{8}{2^4}\right)z^3 + \dots$$

(iii). We have $|z| > 3$. Using this (1) $\Rightarrow$

$$f(z) = 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{z}\left(1+\frac{3}{z}\right)^{-1}$$

$$= 1 + \frac{3}{z}\left(1-\frac{2}{z}+\frac{2^2}{z^2}-\frac{2^3}{z^3}+\dots\right) - \frac{8}{z}\left(1-\frac{3}{z}+\frac{3^2}{z^2}-\frac{3^3}{z^3}+\dots\right)$$

$$= 1 + (3-8)\frac{1}{z} - (6-24)\frac{1}{z^2} + (12-72)\frac{1}{z^3} - \dots$$

$$= 1 - \frac{5}{z} + \frac{18}{z^2} - \frac{60}{z^3} + \dots$$

Example 234 : Expand the function

$f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)}$ for the following regions :

(i). $1 < |z| < 4$; (ii). $|z| < 1$; (iii). $|z| > 4$.

R. U. M. SC. P. '86; D. U. M. SC. P. '88.

Solution : We have $f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)}$

$$= 1 - \frac{1}{z+1} - \frac{4}{z+4} \quad \dots (1).$$

(i). We have $1 < |z| < 4 \Rightarrow |z| > 1$ and $4 > |z|$ ... (2). Now using (2), (1) $\Rightarrow f(z) = 1 - \frac{1}{z}\left(1+\frac{1}{z}\right)^{-1} - \frac{4}{4}\left(1+\frac{z}{4}\right)^{-1}$

$$= 1 - \frac{1}{z}\left(1-\frac{1}{z}+\frac{1}{z^2}-\frac{1}{z^3}+\dots\right) - \left(1-\frac{z}{4}+\frac{z^2}{4^2}-\frac{z^3}{4^3}+\dots\right)$$

$$= \left(1-\frac{1}{z}+\frac{1}{z^2}-\frac{1}{z^3}+\frac{1}{z^4}-\dots\right) - \left(1-\frac{z}{4}+\frac{z^2}{4^2}-\frac{z^3}{4^3}+\dots\right)$$

$$= \dots + \frac{1}{z^4} - \frac{1}{z^3} + \frac{1}{z^2} - \frac{1}{z} + \frac{z}{4} - \frac{z^2}{4^2} + \frac{z^3}{4^3} - \dots$$

(ii). We have $|z| < 1$. Now using this condition

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CHAPTER – 7

RESIDUES AND THE RESIDUE THEOREMS

227. Residue : **D. U. H. '90; D. U. H. T. '83, '89.**

If the function $f(z)$ has an isolated singularity at the point $z = a$, then the coefficient a_{-1} of $\frac{1}{z-a}$ in the Laurent's expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{a_{-n}}{(z-a)^n}$$

$$= a_0 + a_1 (z-a) + a_2 (z-a)^2 + \ldots \ldots$$

$$+ \frac{a_{-1}}{z-a} + \frac{a_{-2}}{(z-a)^2} + \frac{a_{-3}}{(z-a)^3} + \ldots \ldots$$

around $z = a$ is called the residue of $f(z)$ at $z = a$ and it is denoted by Res $[f(z), a]$ or Res (a) or a_{-1}.

N. B. In this book, by $(a_p)_{-1}$ we will mean the residue at the finite point $z = a_p$.

Theorem -200 : If $f(z)$ is analytic inside and on a simple closed curve C except at the point $z = a$ inside C, then

$$a_{-1} = \frac{1}{2\pi i} \oint_C f(z)\, dz.$$

Proof : The Laurent series about $z = a$ is

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n \ldots (1) \text{ where } a_n = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}}\, dz.$$

$n = 0, \pm 1, \pm 2, \ldots (2)$

Now putting $n = -1$ in (2), we get

$a_{-1} = \frac{1}{2\pi i} \oint_C f(z)\, dz$ and the theorem is proved.

Theorem -201 : If $f(z)$ has a simple pole at $z = a$, then show that

$a_{-1} = \lim_{z \to a} (z-a) f(z)$, where a_{-1} is the residue of $f(z)$ at the point $z = a$.

Proof : If $f(z)$ has a simple pole at $z = a$, then the corresponding Laurent series of $f(z)$ is

$f(z) = \frac{a_{-1}}{z-a} + \sum_{n=0}^{\infty} a_n (z-a)^n \ldots (1)$ where $a_{-1} \neq 0$. Now multiplying (1) by $z - a$, then we have

$$(z-a) f(z) = a_{-1} + (z-a) \sum_{n=0}^{\infty} a_n (z-a)^n \ldots (2).$$

Now on letting $z = a$ in (2), we have $a_{-1} = \lim_{z \to a} (z-a) f(z)$ and the theorem is proved.

Theorem -202 : The residue at a finite point $z = a$ is given by the equation Res $[f(z), a] = a_{-1}$ where $f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n \ldots (1)$ is a Laurent expansion of $f(z)$ at $z = a$.

Proof : We have Res $[f(z), a] = \frac{1}{2\pi i} \oint_C f(z)\, dz \ldots (2)$

where a is finite. Let C be a circle $|z - a| = r$ in the region $\Re$ enclosing no singuarity other than a. Than the Laurent series

$z \to a$

Theorem – 205 : If $f(z) = \dfrac{P(z)}{Q(z)}$ where $P(z)$ and $Q(z)$ are analytic at $z = a$ but $P(a) \neq 0$, then $a_{-1} = \dfrac{6P'(a)\,Q''(a) - 2P(a)\,Q'''(a)}{3\,[\,Q''(a)\,]^2}$.

where $P(a)$, $P'(a)$, $Q''(a)$ and $Q'''(a)$ are exist and $Q(z)$ has a double zero at $z = a$.

Proof : Try yourself.

228. Residue at a multiple point :

Theorem – 206 : If $f(z)$ is analytic inside and on a simple closed curve C except at pole $z = a$ of order m inside C, then the residue of $f(z)$ at $z = a$ is

$$a_{-1} = \lim_{z \to a} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \left\{ (z-a)^m f(z) \right\}.$$

D. U. H. T. '84, '86; D. U. M. SC. P. T. '88; D. U. '79; D. U. H. '88. '90; J. U. H.

Proof : (Method 1) : Since $f(z)$ has a pole of order m at z = a, then $f(a) = \frac{F(z)}{(z-a)^m}$, where F(z) is analytic inside and on C and also $F(a) \neq 0$. Then by the Cauchy's (m − 1)th differential integral formula, we have

$$\frac{1}{2\pi i}\oint_C f(z)\,dz$$

$$= \frac{1}{2\pi i}\oint_C \frac{F(z)}{(z-a)^m}\,dz = \frac{1}{2\pi i}\,\frac{F^{(m-1)}(a)}{(m-1)\,!}$$

$$= \lim_{z\to a} \frac{1}{(m-1)\,!}\,\frac{d^{m-1}}{dz^{m-1}}\left\{(z-a)^m f(z)\right\} = a_{-1}$$ and the

answer line

theorem is proved.

and the theorem is proved.

Example 249. Evaluate the residues of $f(z) = \dfrac{z^2}{z^2 + a^2}$ at the poles. $\longrightarrow$ poles.

Solution : The poles of $f(z)$ are given by $z^2 + a^2 = 0$

$\Rightarrow z = \pm\, ai$ and they are simple poles.

Method 1 : Now $\text{Res}\,(ai) = \lim\limits_{z \to ai} \{(z - ai)\, f(z)\}$

$$= \lim_{z \to ai} \left\{(z - ai)\, \frac{z^2}{z^2 + a^2}\right\} = \lim_{z \to ai} \left\{ (z - ai)\, \frac{z^2}{(z - ai)\,(z + ai)} \right\}$$

$$= \lim_{z \to ai} \frac{z^2}{z + ai} = \frac{(ai)^2}{2ai} = \frac{1}{2} ai.$$

Similarly, Res $(-ai) = -\frac{1}{2} ai$

$P(z)$ where $P(z) = z^2$ and $Q(z)$

$\Rightarrow \text{Res}\,(i) = \frac{\quad}{16}$

B

Example-252 : Find the residues of the function $f(z) = \frac{z^2 - 2z}{(z+1)^2(z^2+4)}$ D. U. H. T. '89; D. U. M. SC. P. '88.

Solution : The poles of $f(z)$ are obtained by solving $(z+1)^2(z^2+4) = 0 \Rightarrow f(z)$ has a double pole at $z = -1$ and simple poles at $z = \pm 2i$. Now the residue at $z = -1$ is

$$\lim_{z \to -1} \frac{1}{1!} \frac{d}{dz}\left\{(z+1)^2 \frac{z^2 - 2z}{(z+1)^2(z^2+4)}\right\} = \lim_{z \to -1} \frac{d}{dz}\left\{\frac{z^2-2z}{z^2+4}\right\}$$

$$= \lim_{z \to -1} \frac{z^2-2z}{z^2+4}\left[\frac{2z-2}{z^2-2z} - \frac{2z}{z^2+4}\right] = \frac{3}{5}\left(\frac{-4}{3} + \frac{2}{5}\right) = -\frac{14}{25}$$

Residue at $z = 2i$ is $\lim_{z \to 2i}\left\{(z - 2i)\frac{z^2 - 2z}{(z+1)^2(z^2+4)}\right\}$

$$= \frac{(2i)^2 - 4i}{(2i+1)^2(4i)} = \frac{-4-4i}{(4i^2+4i+1)(4i)} = \frac{1+i}{4+3i} = \frac{7+i}{25}$$

Similarly, residue at $z = -2i$ is $\frac{7-i}{25}$.

Example-253 : Show that $\text{Res}\left[\frac{1}{z^2(z-1)}, 0\right] = -1$.

Solution : We have

$$\frac{1}{z^2(z-1)} = -\frac{1}{z^2}(1-z)^{-1} = -\frac{1}{z^2}(1 + z + z^2 + z^3 + \ldots),$$

$$0 < |z| < 1$$

$$= -\frac{1}{z^2} - \frac{1}{z} - 1 - z - z^2 - \ldots\ldots$$

Solution : Try yourself.

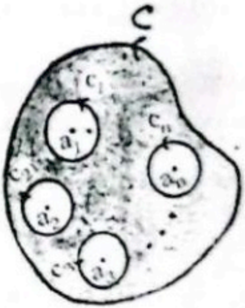
Cauchy's residue theorem - 208 : If $f(z)$ is analytic inside and on a simple closed curve C except at a finite number of n singular points $a_1, a_2, \dots, a_n$ inside C, then

$$\oint_C f(z)\, dz$$

$$= 2\pi i \,[\, \text{Res}\,(a_1) + \text{Res}\,(a_2) + \dots + \text{Res}\,(a_n)],$$ i. e. $2\pi i$ times the sum of the residues at the singularities within C.

R. U. M. SC. P. '85; D. U. M. SC. P. '88; R. U. '76, '82; C. U. H. '81, '88; J. U. H. '87, '91; C. U. M. SC. P. '87; D. U. H. T. '82, '84; D. U. '72.

Proof : Let $C_1, C_2, \dots, C_n$ be n circles with centres at the points $a_1, a_2, \dots, a_n$. Let the radii of these circles are so small that they lie entirely inside C and do not overlap. Then $f(z)$ is analytic in the region between C and these circles and so by a corrolary of the Cauchy's theorem, we have



Example 258 : Show that $\oint_C \frac{e^{tz}}{(z^2+1)^2} dz$

$= \pi i (\sin t - t\cos t)$

where C is the circle $|z| = 3$ and $t > 0$. R. U. H. '80, '82.

Solution : Here the poles of $\frac{e^{tz}}{(z^2+1)^2} = \frac{e^{tz}}{(z-i)^2 (z+i)^2}$

are at $z = \pm i$ inside C and both is of order two.

Now the residue at $z = i$ is $\lim_{z \to i} \frac{1}{1!} \frac{d}{dz} \left\{ \frac{(z-i)^2 e^{tz}}{(z-i)^2 (z+i)^2} \right\}$

Final

$$= \lim_{z \to i} \frac{d}{dz}\left\{\frac{e^{tz}}{(z+i)^2}\right\} = \lim_{z \to i} \frac{t(z+i)^2\, e^{tz} - 2(z+i)\, e^{tz}}{(z+i)^4}$$

$$= \lim_{z \to i} \frac{t(z+i)e^{tz} - 2e^{tz}}{(z+i)^3} = \frac{2(it-1)e^{it}}{(2i)^3} = \frac{-(t+i)e^{it}}{4}$$

Similarly, the residue at $z = -i$ is $\frac{-(t-i)e^{-it}}{4}$

Then by the Cauchy's residue theorem, $\oint_C \frac{e^{tz}}{(z^2+1)^2}\, dz$

$= 2\pi i$ {sum of the residues} $= 2\pi i \left\{\frac{-(t+i)e^{it}}{4} + \frac{-(t-i)e^{-it}}{4}\right\}$

$$= -\frac{\pi i}{2}\left\{t(e^{it} + e^{-it}) + i\,(e^{it} - e^{-it})\right\} = \boxed{\pi i(\sin t - t\cos t).}$$

CONTOUR INTEGRATION

In this chapter we will evalute a varity types of real definite integrals with the help of the Cauchy's residue theorem using a suitable types of closed path or contour. For this reason, the process is called contour integration. Now we will discuss it dividing several forms.

234. (Form 1). Integrals of the form :

$\int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta$, **where** $f(\cos\theta, \sin\theta)$ **is a rational function of** $\cos\theta$ **and** $\sin\theta$ **:**

Final

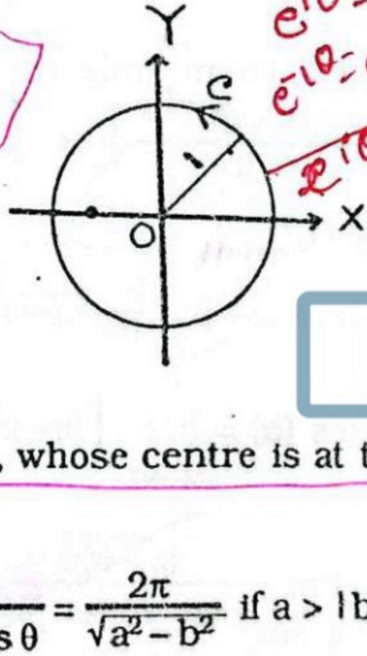
Let $I = \int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta$ ... (1), where $f(\cos\theta, \sin\theta)$ is a rational function of $\cos\theta$ and $\sin\theta$. Let $z = e^{i\theta}$, then $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z^2+1}{2z}$,

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{-i\theta} = \cos\theta - i\sin\theta$

$e^{i\theta} + e^{-i\theta} = 2\cos\theta$

$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{z^2-1}{2iz}$ and $dz = ie^{i\theta}\, d\theta = iz\, d\theta$ or $d\theta = \frac{dz}{iz}$. Now using these (1) $\Rightarrow I = \oint_C g(z)\, dz$,



Final

where C is the unit circle $|z| = 1$, whose centre is at the origin and radius is equal to 1.

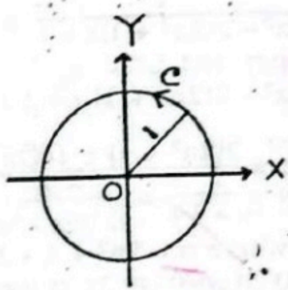
Example 270 : Show that $\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}$ if $a > |b|$.

D. U. H. T. '75, 77, 87; D. U. H. '87.

$e^{-i\theta} = \frac{1}{e^{i\theta}} = \frac{1}{z}$

Example – 276 : Show that $\displaystyle\int_0^{2\pi} \frac{d\theta}{3-2\cos\theta+\sin\theta} = \pi.$

Solution : Suppose $I = \displaystyle\int_0^{2\pi} \frac{d\theta}{3-2\cos\theta+\sin\theta}$ ... (1).



Let $z = e^{i\theta}$, then $\sin\theta = \dfrac{e^{i\theta}-e^{-i\theta}}{2}$

$= \dfrac{z^2-1}{2iz}$, $\cos\theta = \dfrac{e^{i\theta}+e^{-i\theta}}{2}$

$= \dfrac{z^2+1}{2z}$ and $dz = ie^{i\theta}\,d\theta = iz\,d\theta$ or

$d\theta = \dfrac{dz}{iz}$. Now using these (1) $\Rightarrow I =$

$\displaystyle\oint_C \frac{2dz}{(1-2i)z^2+6iz-1-2i}$ ...(2), where C is the circle $|z| = 1$.

The poles of $\dfrac{2}{(1-2i)\,z^2+6iz-1-2i}$ are obtained by solving $(1-2i)z^2+6iz-1-2i = 0 \Rightarrow z = \dfrac{-6i \pm 4i}{2(1-2i)}$

$= \dfrac{-2i}{10}(1+2i), \dfrac{-10i}{10}(1+2i) \Rightarrow z = \dfrac{1}{5}(2-i)$, $2-i$ and they are simple. Of them only the pole $\dfrac{1}{5}(2-i)$ lies inside C. Now

Res $\left\{\dfrac{1}{5}(2-i)\right\}$

$= \displaystyle\lim_{z\to\frac{1}{5}(2-i)} \left[\left\{z-\frac{1}{5}(2-i)\right\}\frac{2}{(1-2i)z^2+6iz-1-2i}\right]$

[by L. Hospital's rule]

$$= \lim_{z \to \frac{1}{5}(2-i)} \frac{2}{2(1-2i)\,z + 6i}$$

$$= \frac{5}{2-2-i-4i+15i} = \frac{1}{2i}$$

Now by the Cauchy's residue theorem (2) $\Rightarrow$

$I = 2\pi i.\ \text{Res}\left\{\frac{1}{5}(2-i)\right\} = 2\pi i.\ \frac{1}{2i} = \pi$ and the required value is obtained.

235. (Form 2). Integrals of the form : $\int_{-\infty}^{\infty} f(x)\,dx$, **where** $f(x)$ **is a rational function of** x **:** Consider $\oint_C f(z)\,dz$, where C is the contour consisting of :

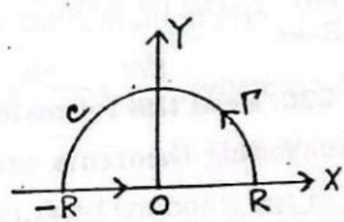
(i). the x- axis from $-R$ to R, where R is large;

(ii). the upper semi-circle Γ of the circle $|z| = R$, which lies above the x–axis.

Then let $R \to \infty$ and if $f(x)$ is an even function this can be used to evalute the integral $\int_0^{\infty} f(x)\,dx$.

Theorem – 216 : If $|f(z)| \le \frac{M}{R^k}$ for $z = Re^{i\theta}$, where $k > 1$ and M is a constant, then show that $\lim\limits_{R \to \infty} \int_\Gamma f(z)\,dz = 0$, where Γ is the upper semi–circle of radius R.

Proof : We have $\frac{M}{R^k}$ is the upper bound of $|f(z)|$ and πR is the upper semi circular arc length of Γ, then



$$\left| \int_\Gamma f(z)\,dz \right| \le \left(\frac{M}{R^k} \right) (\pi R)$$

$$= \frac{\pi M}{R^{k-1}} \ldots (1).$$

Now taking $R \to \infty$ in both sides of (1)

$$\Rightarrow \lim_{R \to \infty} \left| \int_\Gamma f(z)\,dz \right| = 0 \Rightarrow \lim_{R \to \infty} \int_\Gamma f(z)\,dz = 0$$ and the theorem is proved.

Example - 289 : Show that $\int_0^\infty \frac{dx}{x^4+1} = \frac{\pi}{2\sqrt{2}}$.

D. U. '84; D. U. H. S. T. '85; D. U. M. SC. P. 88.

Solution : Try yourself or, see the above example.

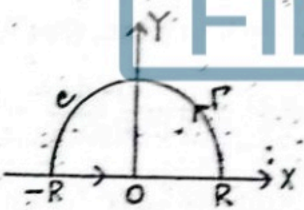
Example - 290 : Show that $\int_0^\infty \frac{dx}{x^6+1} = \frac{\pi}{3}$.

D. U. H. T. '86, '88.

Solution : Consider $\oint_C f(z)\, dz$ where $f(z) = \frac{1}{z^6+1}$ and C is the contour consisting of :

(i). the x-axis from $-R$ to R, where R is large;

(ii). the upper semi-circle Γ of the circle $|z| = R$, which lies above the x-axis.



Now the poles of $f(z)$ are given by $z^6 + 1 = 0 \Rightarrow z^6 = -1 = e^{(2n+1)\pi i} \Rightarrow z = e^{(2n+1)\pi i/6}$, where $n = 0, 1, 2, 3, 4, 5$

$\Rightarrow z = e^{\pi i/6}, e^{\pi i/2}, e^{5\pi i/6}, e^{7\pi i/6}, e^{9\pi i/6}, e^{11\pi i/6}$ and they are simple. Of them only the poles $z = e^{\pi i/6}, e^{\pi i/2}$ and $e^{5\pi i/6}$ lie within C.

Now using L' Hospital's rule, the residue at the pole $z = \alpha = \lim_{z\to\alpha} (z-\alpha) f(z) = \lim_{z\to\alpha} \frac{z-\alpha}{z^6+1} = \lim_{z\to\alpha} \frac{1}{6z^5}$

$$= \frac{1}{6\alpha^5} = \begin{cases} \frac{1}{6} e^{-5\pi i/6} = \frac{1}{6} e^{-\pi i} e^{\pi i/6} = -\frac{1}{6} e^{\pi i/6} \text{ at } z = e^{\pi i/6} \\ \frac{1}{6} e^{-5\pi i/2} = \frac{1}{6} e^{-2\pi i} e^{-\pi i/2} = -\frac{i}{6} \quad \text{at } z = e^{\pi i/2} \\ \frac{1}{6} e^{-25\pi i/6} = \frac{1}{6} e^{-4\pi i} e^{-\pi i/6} = \frac{1}{6} e^{-\pi i/6} \text{ at } z = e^{5\pi i/6} \end{cases}$$

Now by the Cauchy's residue theorem, we have

$$\oint_C f(z)\,dz = \int_R^R f(x)\,dx + \int_\Gamma f(z)\,dz$$

$= 2\pi i$ (sum of residues)

$$= 2\pi i\left(-\frac{1}{6}e^{\pi i/6} - \frac{i}{6} + \frac{1}{6}e^{-\pi i/6}\right) = \frac{-\pi i}{3}\left(e^{\pi i/6} - e^{-\pi i/6} + i\right)$$

$= (-\frac{\pi i}{3})\,(2i\sin\frac{\pi}{6} + i) = \frac{\pi}{3}(1 + 1) = \frac{2\pi}{3}$ ... (1) . Here

$\lim\limits_{z\to\infty} z\,f(z) = \lim\limits_{z\to\infty}\frac{z}{z^6+1} = 0$, then $\lim\limits_{R\to\infty}\int_\Gamma f(z)\,dz = 0$ ...

(2). Now taking limit $R\to\infty$ in (1) and using (2)

$\Rightarrow \int_{-\infty}^{\infty} f(x)\,dx = \frac{2\pi}{3} \Rightarrow 2\int_0^\infty \frac{dx}{x^6+1} = \frac{2\pi}{3} \Rightarrow \int_0^\infty \frac{dx}{x^6+1} = \frac{\pi}{3}$ and

the result is proved.

237. (Form 3). Integrals of the form : $\int_{-\infty}^{\infty} f(x) \cos mx\, dx$ **or** $\int_{-\infty}^{\infty} f(x) \sin mx\, dx$ **where** $f(x)$ **is a rational function of** x **:** Consider $\oint_C e^{imz} f(z)\, dz$, $m > 0$ where C is the contour consisting of :

(i). the x-axis from $-R$ to R, where R is large;

(ii). the upper semi-circle Γ of the circle $|z| = R$, which lies above the x-axis.

Then let $R \to \infty$ and if $f(x)$ is an even function, this can be used to evalute the integral $\int_{-\infty}^{\infty} e^{imx} f(x)\, dx$, $m > 0$.

238. Jordan's inequality : The inequality $\frac{2}{\pi} \leq \frac{\sin \theta}{\theta} \leq 1$ or $\frac{2\theta}{\pi} \leq \sin \theta \leq \theta$ is called the Jordan's inequality.

where $0 \leq \theta \leq \frac{\pi}{2}$.

Jordan lemma (Theorem —221) : If $|f(z)| \leq \frac{M}{R^k}$ for $z = Re^{i\theta}$, where $k > 0$ and M is a constant.

then $\lim_{R \to \infty} \int_\Gamma e^{imz} f(z)\, dz = 0$, where Γ is the upper semi-circle of radius R and m is a positive constant.

Proof : If $z = Re^{i\theta}$, then $\int_\Gamma e^{imz} f(z)\, dz$

$$= \int_0^\pi e^{imRe^{i\theta}} f(Re^{i\theta})\, d(Re^{i\theta}) = i\int_0^\pi e^{imRe^{i\theta}} f(Re^{i\theta})\, Re^{i\theta}\, d\theta$$

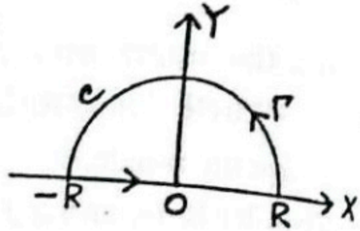
$$\Rightarrow \left| \int_\Gamma e^{imz} f(z)\, dz \right| = \left| \int_0^\pi e^{imRe^{i\theta}} f(Re^{i\theta})\, Re^{i\theta}\, d\theta \right|$$

$$\leq \int_0^{\pi} \left| e^{imRe^{i\theta}} f(Re^{i\theta})\, Re^{i\theta}\, d\theta \right| = \int_0^{\pi} e^{-mR\sin\theta} \left| f(Re^{i\theta}) \right| R d\theta$$

$$\leq \frac{M}{R^{k-1}} \int_0^{\pi} e^{-mR\sin\theta}\, d\theta$$

$$= \frac{2M}{R^{k-1}} \int_0^{2\pi} e^{-mR\sin\theta}\, d\theta \ldots (1).$$

But we have $\sin\theta \geq 2\theta/\pi$. (2)
for $0 \leq \theta \leq \pi/2$. Using (2).



$$(1) \Rightarrow \left| \int_{\Gamma} e^{imz} f(z) dz \right|$$

$$\leq \frac{2M}{R^{k-1}} \int_0^{\pi/2} e^{-2mR\theta/\pi}\, d\theta$$

$$= \frac{2M}{R^{k-1}} \left[\frac{\pi e^{-2mR\theta/\pi}}{-2MR} \right]_0^{\pi/2} = \frac{\pi M}{mR^k} (1 - e^{-mR}) \ldots (3)$$

Now taking $R \to \infty$ in both sides of (3) $\Rightarrow$

$$\lim_{R\to\infty} \left| \int_{\Gamma} e^{imz} f(z)\, dz \right| = 0 \Rightarrow \lim_{R\to\infty} \int_{\Gamma} e^{imz} f(z)\, dz = 0$$

and the theorem is proved.

N. B. If $m<0$, then also $\lim\limits_{R\to\infty} \int_{\Gamma} e^{imz} f(z)\, dz = 0$ where Γ is the lower semi-circle of radius R and it can be proved by changing $z = -w$ in the above theorem.

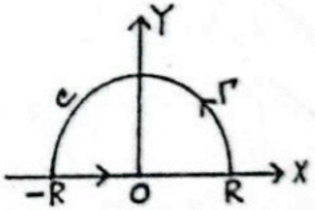
Example 296 : Show that $\int_0^\infty \frac{\cos mx}{x^2 + a^2}\,dx = \frac{\pi}{2a} e^{-ma}$.

$m \geq 0$ and $a > 0$. **D. U. H. '78, '88; R. U. H. '76.**

Solution : Consider $\oint_C f(z)\,dz$ where $f(z) = \frac{e^{imz}}{z^2 + a^2}$ and C is the contour consisting of :

(i). the x-axis from –R to R, where R is large.

(ii). the upper semi-circle Γ of the circle $|z| = R$ which lies above the x-axis.



The poles of $f(z) = \dfrac{e^{imz}}{z^2+a^2}$ are obtained by solving

$z^2 + a^2 = 0 \Rightarrow z = \pm\, ai$ and they are simple poles. Of which only $z = ai$ lies inside C.

$$\text{Res}\,[f(z), ai] = \lim_{z\to ai}\left\{(z-ai)\frac{e^{imz}}{(z-ai)(z+ai)}\right\} = \frac{e^{-ma}}{2ai}.$$

Now by the Cauchy's residue theorem, we have

$$\oint_C f(z)\,dz = \int_{-R}^{R} f(x)\,dz + \int_{\Gamma} f(z)\,dz$$

$$= 2\pi i\,\text{Res}\,[f(z), ai] = 2\pi i\left(\frac{e^{-ma}}{2ai}\right) = \frac{\pi e^{-ma}}{a} \ldots (1).$$

Here $\lim\limits_{z\to\infty} \dfrac{1}{z^2+a^2} = 0$ and $f(z)$ is a function of the form $e^{imz}F(z)$, where $F(z) = \dfrac{1}{z^2+a^2}$ and then by the Jordan's lemma, $\lim\limits_{R\to\infty} \int_{\Gamma} f(z)\,dz = 0 \ldots (2)$. Now taking limit $R\to\infty$ in (1) and using (2) $\Rightarrow \int_{-\infty}^{\infty} f(x)\,dx = \dfrac{\pi e^{-ma}}{a} \Rightarrow 2\int_0^{\infty} f(x)\,dx = \dfrac{\pi e^{-ma}}{a}$.

$\Rightarrow \int_0^{\infty} \dfrac{\cos mx}{x^2+a^2} = \dfrac{\pi e^{-ma}}{2a}$ and the required result is obtained.

Example – 297 : Show that (a) $\int_0^{\infty} \dfrac{\cos x}{1+x^2}\,dx = \dfrac{\pi}{2e}$;

(b) $\int_0^{\infty} \dfrac{\cos mx}{x^2+1} = \dfrac{\pi}{2}$; R. U. M. SC. P. '84.

(c) $\int_0^{\infty} \dfrac{\cos x}{x^2+a^2} = \dfrac{\pi e^{-a}}{2a}$, $a > 0$.

Final

Solution : Use the above example.

Example – 298 : Using the above example, show that $\int_0^{\infty} \dfrac{x \sin mx}{x^2+a^2}\,dz = \dfrac{\pi}{2}e^{-ma}$, where $m \ge 0$ and $a > 0$.

Solution : By the above example, we have $\int_0^\infty \frac{\cos mx}{x^2+a^2}\,dx = \frac{\pi}{2a} e^{-ma}$ ... (1) where $m \geq 0$ and $a > 0$.

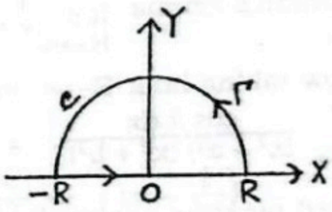
Now differentiating both sides of (1), with respect to m, we have $\int_0^\infty \frac{-x \sin mx}{x^2+a^2}\,dx = -\frac{a\pi}{2a} e^{-ma}$

$\Rightarrow \int_0^\infty \frac{x \sin mx}{x^2+a^2}\,dx = \frac{\pi}{2} e^{-ma}$ and the required result is obtained.

Example 299 : Show that $\int_{-\infty}^{\infty} \frac{\cos x\,dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{a^2-b^2}\left(\frac{e^{-b}}{b}-\frac{e^{-a}}{a}\right)$ where $a > b > 0$.

D. U. H. '89, '90; D. U. H. T. '77, 82, 87.

Solution : Consider $\oint_C f(z)\,dz$ where $f(z) = \frac{e^{iz}}{(z^2+a^2)(z^2+b^2)}$ and C is the contour consisting of : (i) the x-axis from $-R$ to R, where R is large; (ii) the upper semi-circle Γ of the circle $|z| = R$ which lies above the x-axis.



The poles of $f(z) = \frac{e^{iz}}{(z^2+a^2)(z^2+b^2)}$ are obtained by solving $(z^2+a^2)(z^2+b^2) = 0 \Rightarrow z = \pm ai, \pm bi$ which are simple poles. Only the poles $z = ai$ and $z = bi$ lie inside C.

. Now Res (ai) = $\lim_{z \to ai} \left\{ (z - ai) \frac{e^{iz}}{(z - ai)(z + ai)(z^2 + b^2)} \right\}$

$= \lim_{z \to ai} \frac{e^{iz}}{(z + ai)(z^2 + b^2)} = \frac{e^{-a}}{2ai(b^2 - a^2)} = -\frac{e^{-a}}{2ai(a^2 - b^2)}$.

Again Res (bi) = $\frac{e^{-b}}{2bi(a^2 - b^2)}$.

Then by the Cauchy's residue theorem, we have

$\oint_C f(z)\, dz = \int_{-R}^{R} f(x)\, dx + \int_\Gamma f(z)\, dz$

$= 2\pi i\,[\text{Res}\,(ai) + \text{Res}\,(bi)\,]$

$= 2\pi i \left[-\frac{e^{-a}}{2ai(a^2 - b^2)} + \frac{e^{-b}}{2bi(a^2 - b^2)} \right] = \frac{\pi}{a^2 - b^2} \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right)$

... (1)

Here $\lim_{z \to \infty} \frac{1}{(z^2 + a^2)(z^2 + b^2)} = 0$ and $f(z)$ is a function of the form $e^{imz}\, F(z)$, where $F(z) = \frac{1}{(z^2 + a^2)(z^2 + b^2)}$ and then by the Jordan's lemma $\lim_{R \to \infty} \int_\Gamma f(z)\, dz = 0$ ... (2).

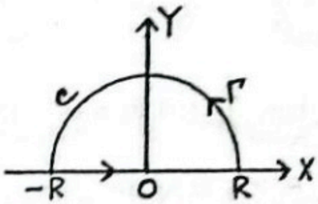
Now taking limit $R \to \infty$ in (1) and using (2) $\Rightarrow$

$\int_{-\infty}^{\infty} \frac{\cos x\, dx}{(x^2 + a^2)(x^2 + b^2)} = \frac{\pi}{a^2 - b^2} \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right)$ and the required result is obtained.

Example –301 : Show that $\int_0^\infty \frac{\cos mx}{(x^2+a^2)^2}\,dx = \frac{\pi(1+ma)e^{-ma}}{4a^3}$.

where $a > 0$ and $m \geq 0$. **D. U. H. T. '86; C. U. '81.**

Solution : Consider $\oint_C f(z)\,dz$ where $f(z) = \frac{e^{imz}}{(z^2+a^2)^2}$ and C is the contour consisting of : **(i)** the x-axis from $-R$ to R, where R is large; **(ii)** the upper semi-circle Γ of the circle $|z| = R$ which lies above the x-axis.



The poles of $f(z) = \frac{e^{imz}}{(z^2+a^2)^2}$ are obtained by solving $(z^2+a^2)^2 = 0 \Rightarrow z = \pm\, ai$ which are of order 2. Only the pole $z = ai$ lies inside C. Now Res (ai) =

$$\lim_{z \to ai} \frac{d}{dz}\left\{(z-ai)^2 \frac{e^{imz}}{(z-ai)^2\,(z+ai)^2}\right\}$$

$$= \lim_{z \to ai} \frac{d}{dz}\left\{\frac{e^{imz}}{(z+ai)^2}\right\}$$

$$= \lim_{z \to ai}\left[\frac{e^{imz}}{(z+ai)^2}\left\{im - \frac{2}{z+ai}\right\}\right] = \frac{e^{-ma}}{(2ai)^2}\left\{im - \frac{2}{2ai}\right\}$$

$$= -\frac{e^{-ma}}{4a^2}\left(\frac{-am-1}{ai}\right) = \frac{e^{-ma}\,(ma+1)}{4a^3 i}.$$ Then by the Cauchy's residue theorem, we have $\oint_C f(z)\,dz$

$$= \int_{-R}^{R} f(x)\,dx + \int_\Gamma f(z)\,dz = 2\pi i \text{ Res } (ai)$$

$$= 2\pi i \left\{ \frac{e^{-ma}(ma+1)}{4a^3 i} \right\} = \frac{\pi(1+ma)e^{-ma}}{2a^3} \ldots (1).$$ Here

$\lim\limits_{z\to\infty} \frac{1}{(z^2+a^2)^2} = 0$ and $f(z)$ is a function of the form

e^{imz} F(z). where $F(z) = \frac{1}{(z^2+a^2)^2}$ and then by the Jordan's

lemma

$$\lim_{R\to\infty} \int_{\Gamma} f(z)\, dz = 0 \ldots (2).$$

Now taking limit $R\to\infty$ in (1) and using (2) $\Rightarrow \int_{-\infty}^{\infty} f(x)\, dx$

$$= \frac{\pi(1+ma)e^{-ma}}{2a^3} \Rightarrow \int_{-\infty}^{\infty} \frac{\cos mx}{(x^2+a^2)^2}\, dx = \frac{\pi(1+ma)e^{-ma}}{2a^3}$$

$$\Rightarrow 2\int_0^{\infty} \frac{\cos mx}{(x^2+a^2)^2}\, dx = \frac{\pi(1+ma)e^{-ma}}{2a^3} \Rightarrow \int_0^{\infty} \frac{\cos mx}{(x^2+a^2)^2}\, dx$$

$= \frac{\pi(1+ma)e^{-ma}}{4a^3}$ and the required result is proved.